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// BootRunner.cs
using System.Collections;
using Sy.Core;
using UnityEngine;
using UnityEngine.SceneManagement;
using Logger = Sy.Core.Logger;
namespace Sy.CrazyHospital.Boot
{
    public static class SceneIds
    {
        public const string Boot = "S00_Boot";
        public const string Main = "S09_Main";
    }
    /// <summary>
    /// S00_Boot scene host: register mock services -> run seven steps -> switch to main scene.
    /// On failure it stays on S00_Boot and logs the error (no dead-end while loops).
    /// </summary>
    public sealed class BootRunner : MonoBehaviour
    {
        IEnumerator Start()
        {
            BootServices.RegisterAll();
            var sequence = new BootSequence(SwitchToMain);
            if (!sequence.RunAll())
            {
                Logger.Error("BOOT", "boot sequence failed - stay on S00_Boot");
                yield break;
            }
            yield return null;
        }
        bool SwitchToMain()
        {
            StartCoroutine(LoadMainRoutine());
            return true;
        }
        IEnumerator LoadMainRoutine()
        {
            var op = SceneManager.LoadSceneAsync(SceneIds.Main, LoadSceneMode.Single);
            while (op != null && !op.isDone) yield return null;
        }
    }
}

// BootSequence.cs
using System;
using System.Collections.Generic;
using Sy.Core;
using Sy.Platform;
namespace Sy.CrazyHospital.Boot
{
    /// <summary>
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/// Seven-step boot sequence: Logger -&gt; Time -&gt; Config -&gt; Save -&gt; Locale -&gt; Ads.Preload -&gt; SwitchM
/// each step logs "[BOOT] n/7 name ok|FAIL". A step exception never blocks later steps.
/// &lt;/summary&gt;
public sealed class BootSequence
{
    public const int TotalSteps = 7;
    readonly Func&lt;bool&gt; _switchScene;
    readonly List&lt;string&gt; _lines = new();
    public BootSequence(Func&lt;bool&gt; switchScene)
    {
        _switchScene = switchScene;
    }
    public IReadOnlyList&lt;string&gt; Lines =&gt; _lines;
    public bool RunAll()
    {
        bool ok = true;
        ok &= Run(1, "Logger", LoggerStep);
        ok &= Run(2, "Time", TimeStep);
        ok &= Run(3, "Config", ConfigStep);
        ok &= Run(4, "Save", SaveStep);
        ok &= Run(5, "ILocale", LocaleStep);
        ok &= Run(6, "Ads.Preload", AdsStep);
        ok &= Run(7, "SwitchMain", () =&gt; _switchScene != null && _switchScene());
        return ok;
    }
    bool Run(int n, string name, Func&lt;bool&gt; step)
    {
        bool result;
        try
        {
            result = step();
        }
        catch (Exception e)
        {
            Logger.Error("BOOT", $"step {n} {name} exception: {e.Message}");
            result = false;
        }
        string line = $"[BOOT] {n}/{TotalSteps} {name} {(result ? "ok" : "FAIL")}";
        _lines.Add(line);
        Logger.Info("BOOT", line);
        return result;
    }
    static bool LoggerStep() =&gt; true; // static facade self-initializes on first call
    static bool TimeStep()
    {
        var time = ServiceLocator.Get&lt;ITimeService&gt;();
        return time.ServerNowMs() &gt; 0 && !time.TimeCheated();
    }
    static bool ConfigStep()
    {

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        var config = ServiceLocator.Get<IConfigService>();
        var update = config.CheckUpdate("mock://manifest");
        return update.NewVersion != null; // mock = no hot update
    }
    static bool SaveStep()
    {
        var save = ServiceLocator.Get<ISaveService>();
        save.Save("boot_probe", "{}");
        return !string.IsNullOrEmpty(save.Load("boot_probe"));
    }
    static bool LocaleStep()
    {
        var locale = ServiceLocator.Get<ILocaleService>();
        return !string.IsNullOrEmpty(locale.GetFontSet().NormalFont);
    }
    static bool AdsStep()
    {
        ServiceLocator.Get<IAdsService>().Preload("boot_warmup");
        return true;
    }
}

// BootServices.cs
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using Sy.Core;
using Sy.Platform;
using UnityEngine;
using Logger = Sy.Core.Logger;
namespace Sy.CrazyHospital.Boot
{
    /// <summary>
    /// Dev-time mock service set (SY MOCK scope). All seven boot steps depend on this set.
    /// </summary>
    public static class BootServices
    {
        public static void RegisterAll()
        {
            if (ServiceLocator.TryGet<ITimeService>(out _)) return; // idempotent: double register would throw
            ServiceLocator.Register<ITimeService>(new MockTimeService());
            ServiceLocator.Register<IConfigService>(new MockConfigService());
            ServiceLocator.Register<ISaveService>(new MockSaveService());
            ServiceLocator.Register<ILocaleService>(new MockLocaleService());
            ServiceLocator.Register<IAdsService>(new MockAdsService());
            Logger.Info("BOOT", "mock services registered (SY MOCK)");
        }
    }

    public sealed class MockTimeService : ITimeService
    {

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    public TimeSource Source => TimeSource.Mock;
    public long ClientNowMs() => DateTimeOffset.UtcNow.ToUnixTimeMilliseconds();
    /// <summary>SY MOCK: device time doubles as server time until the real chain lands.</summary>
    public long ServerNowMs() => ClientNowMs();
    public bool TimeCheated() => Math.Abs(ServerNowMs() - ClientNowMs()) > 120_000;
}

public sealed class MockConfigService : IConfigService
{
    public T GetTable<T>(string tableId) where T : class
        => throw new KeyNotFoundException("mock config table not loaded: " + tableId);
    public bool TryGetTable<T>(string tableId, out T table) where T : class
    {
        table = null;
        return false;
    }
    public ConfigUpdateResult CheckUpdate(string manifestUrl)
        => new ConfigUpdateResult(false, "mock-0.0.1");
    public void SwapPolicy(ConfigSwapPolicy policy) { }
    public string PinConfigVersion(string scope) => "mock-0.0.1";
}

public sealed class MockSaveService : ISaveService
{
    const string Prefix = "sy_save_";
    public void Save(string slot, string payloadJson) => PlayerPrefs.SetString(Prefix + slot, payloadJson);
    public string Load(string slot) => PlayerPrefs.GetString(Prefix + slot, string.Empty);
    /// <summary>No cloud: local storage is the authority (zero-server doctrine).</summary>
    public SyncResult CloudSync() => new SyncResult(false, null);
}

public sealed class MockLocaleService : ILocaleService
{
    public void SetLang(string lang) { }
    /// <summary>Missing key falls back to the key itself with a warning.</summary>
    public string GetText(string key, params string[] args)
    {
        Logger.Warn("LOCALE", "missing i18n key (fallback): " + key);
        return key;
    }
    public FontSet GetFontSet()
        => new FontSet("G09Main", "G09Main", "G09Main");
}

public sealed class MockAdsService : IAdsService
{
    public event Action<AdLifecycle> OnAdEvent;
    public Task<RewardResult> ShowRewarded(string pointId)
        => Task.FromResult(new RewardResult { Completed = true, Reason = "mock" });
    public void ShowInterstitial(string pointId) { }
    public void ShowBanner(string pointId) { }
    public void Preload(string pointId)
    {
        Logger.Info("ADS", "preload (mock): " + pointId);
    }
}

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        OnAdEvent?.Invoke(new AdLifecycle(AdLifecycleType.Fill, pointId, "mock"));
    }
}

// HospitalAlbum.cs
// HospitalAlbum.cs - the three-volume album + wonder-drug research
// line (G09-T009, SYS07/SYS08, M07/M08). The album is the collection
// depth face: every absurd disease a patient books, every odd doctor
// hired and every wonder drug researched lands in its frozen volume
// (diseases / doctors / drugs - zero real-medicine vocabulary, every
// row already frozen in HospitalData and HospitalDoctors). The
// collection law is the frozen section-five closed set: a completed
// volume pays income +2% (AlbumBonusPerVolumePct) - the passive face
// the check reconciles; partial volumes only move the rate face. The
// research line (M07) spends gold on the frozen rows and keeps each
// drug's own bonus on the accounting faces (reg_fee_pct /
// bed_fee_pct / cure_rate) - the host folds those into the intake
// economy at the wiring slice, this face keeps the books. Unlocks
// book evt_album_unlock, a purchase books evt_research, a volume
// crossing complete books evt_album_complete once; doctor hires fold
// silently through SyncDoctors (the hire face already booked
// evt_doctor_hire). Second-run inheritance keeps the three volumes
// (InheritAlbumsKept). Pure C#, no MonoBehaviour, zero RNG.
using System;
using System.Collections.Generic;
namespace Sy.CrazyHospital
{
    /// <summary>SYS08: the three-volume album + wonder-drug research
    /// line data face. The host wires RegisterDisease / SyncDoctors /
    /// TryResearch; headless gates drive the static solvers and the
    /// armed faces directly.</summary>
    public sealed class HospitalAlbum
    {
        // ---- the frozen volume set (diseases / doctors / wonder drugs) ----
        public const string DiseaseVolumeId = "album_disease";
        public const string DrugVolumeId = "album_drug";
        public static string DoctorVolumeId => HospitalDoctors.DoctorAlbumId;
        readonly HospitalDoctors _doctors;
        readonly List<string> _unlocked = new List<string>();
        readonly HashSet<string> _completed = new HashSet<string>(); // the once-guard of evt_album_comple
        readonly List<string> _funnel = new List<string>();
        int _researchSpendHc;
        public HospitalAlbum(HospitalDoctors doctors = null) { _doctors = doctors; }
        // ---- live faces ----
        public IReadOnlyList<string> Unlocked => _unlocked;
        public IReadOnlyList<string> Funnel => _funnel;
        public int ResearchSpendHc => _researchSpendHc;
        /// <summary>The three volume ids in pack order (diseases /
        /// doctors / wonder drugs).</summary>
        public static string[] VolumeIds()

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        => new[] { DiseaseVolumeId, HospitalDoctors.DoctorAlbumId, DrugVolumeId };
/// <summary>Every frozen album entry id of one volume, table
/// order (the catalog face).</summary>
public static List<string> EntriesOf(string volumeId)
{
    var ids = new List<string>();
    string vol = volumeId ?? "";
    foreach (var d in HospitalData.Diseases()) if (d.albumId == vol) ids.Add(d.id);
    foreach (var d in HospitalDoctors.Doctors()) if (d.albumId == vol) ids.Add(d.id);
    foreach (var d in HospitalData.Research()) if (d.albumId == vol) ids.Add(d.id);
    return ids;
}
/// <summary>The volume an entry id belongs to (" = not a frozen
/// album entry).</summary>
public static string VolumeOf(string entryId)
{
    string id = entryId ?? "";
    foreach (var d in HospitalData.Diseases()) if (d.id == id) return d.albumId;
    foreach (var d in HospitalDoctors.Doctors()) if (d.id == id) return d.albumId;
    foreach (var d in HospitalData.Research()) if (d.id == id) return d.albumId;
    return "";
}
public static int TotalCountOf(string volumeId) => EntriesOf(volumeId).Count;
public int UnlockedCountOf(string volumeId)
{
    string vol = volumeId ?? "";
    int n = 0;
    for (int i = 0; i < _unlocked.Count; i++)
        if (VolumeOf(_unlocked[i]) == vol) n++;
    return n;
}
/// <summary>Collection rate of one volume (floor percent).</summary>
public int RatePctOf(string volumeId)
    => SolveRatePct(UnlockedCountOf(volumeId), TotalCountOf(volumeId));
/// <summary>Pure rate solver (headless face: the percent math
/// has no runtime state).</summary>
public static int SolveRatePct(int unlocked, int total)
    => total <= 0 ? 0 : unlocked * 100 / total;
public bool IsComplete(string volumeId) => RatePctOf(volumeId) >= 100;
public int CompletedVolumeCount
{
    get { int n = 0; foreach (var v in VolumeIds()) if (IsComplete(v)) n++; return n; }
}
/// <summary>The section-five passive face: each completed
/// volume pays income +2% (the frozen closed set).</summary>
public int IncomeBonusPct => CompletedVolumeCount * HospitalData.AlbumBonusPerVolumePct;
/// <summary>Pure bonus solver (headless face).</summary>
public static int SolveBonusPct(int completedVolumes)
    => completedVolumes * HospitalData.AlbumBonusPerVolumePct;
/// <summary>The research accounting face: the sum of bonusValue

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/// over the researched rows of one bonus domain
/// (reg_fee_pct / bed_fee_pct / cure_rate) - the wiring slice
/// folds this into the intake economy.</summary>
public int ResearchBonusOf(string bonusId)
{
    string dom = bonusId ?? "";
    int n = 0;
    foreach (var id in _unlocked)
        foreach (var r in HospitalData.Research())
            if (r.id == id && r.bonusId == dom) n += r.bonusValue;
    return n;
}
/// </summary>Unlocks one album entry (the comedy-events face:
/// the E06 / E09 album unlocks land here). A new unlock books
/// evt_album_unlock; a volume crossing complete books
/// evt_album_complete once.</summary>
public bool Unlock(string entryId)
{
    string vol = VolumeOf(entryId);
    if (vol.Length == 0 || _unlocked.Contains(entryId)) return false;
    _unlocked.Add(entryId);
    _funnel.Add("evt_album_unlock@" + vol + ":" + entryId);
    if (IsComplete(vol) && _completed.Add(vol))
        _funnel.Add("evt_album_complete@" + vol);
    return true;
}
/// </summary>The patient-intake face (M08: the absurd disease
/// registers into the volume the moment its patient books
/// in) - the host calls this from the intake wiring.</summary>
public bool RegisterDisease(string diseaseId) => Unlock(diseaseId);
/// </summary>Folds the hired roster into the doctor volume (the
/// hire face already booked evt_doctor_hire, so the sync rides
/// silent - no duplicate funnel lines).</summary>
public void SyncDoctors()
{
    if (_doctors == null) return;
    foreach (var d in _doctors.Collection)
    {
        string vol = VolumeOf(d.id);
        if (vol.Length == 0 || _unlocked.Contains(d.id)) continue;
        _unlocked.Add(d.id);
        if (IsComplete(vol) && _completed.Add(vol))
            _funnel.Add("evt_album_complete@" + vol);
    }
}
/// </summary>The wonder-drug research line (M07): spends gold on
/// one frozen row. Returns the cost booked (0 = refused:
/// unknown row, already researched, or gold short - nothing
/// changes). The purchase books evt_research, the unlock rides
/// Unlock; the caller books the spent gold into its own ledger
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/// at the host-wiring slice.</summary>
public int TryResearch(string drugId, int goldHc)
{
    foreach (var r in HospitalData.Research())
    {
        if (r.id != drugId) continue;
        if (goldHc < r.cost || !_unlocked.Contains(drugId)) return 0;
        _researchSpendHc += r.cost;
        _funnel.Add("evt_research@" + drugId + ":" + r.cost);
        Unlock(drugId);
        return r.cost;
    }
    return 0;
}

/// </summary>Second-run inheritance (InheritAlbumsKept):
/// re-seeds the unlocked set silently - a fresh run starts
/// with the volumes it already earned (the flex face).</summary>
public void InheritFrom(IReadOnlyList<string> entryIds)
{
    if (entryIds == null) return;
    foreach (var id in entryIds)
    {
        string vol = VolumeOf(id);
        if (vol.Length == 0 || !_unlocked.Contains(id)) continue;
        _unlocked.Add(id);
        if (IsComplete(vol)) _completed.Add(vol);
    }
}

/// </summary>Gate face: headless checks seed entries without
/// funnel lines (the armed state is the precondition, not the
/// story).</summary>
public void ArmForGate(params string[] entryIds) => InheritFrom(entryIds);
/// </summary>Canonical fingerprint: the zero-RNG parity
/// instrument (same script = same bytes).</summary>
public string Fingerprint()
{
    var sb = new System.Text.StringBuilder();
    sb.Append("unlocked=").Append(_unlocked.Count);
    foreach (var v in VolumeIds())
        sb.Append('|').Append(v).Append('=')
            .Append(UnlockedCountOf(v)).Append('/').Append(TotalCountOf(v));
    sb.Append("|complete=").Append(_completed.Count);
    sb.Append("|income=").Append(IncomeBonusPct);
    sb.Append("|spend=").Append(_researchSpendHc);
    sb.Append("|fun=").Append(_funnel.Count);
    return sb.ToString();
}
}

// HospitalAlbumTabs.cs
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// HospitalAlbumTabs.cs - the three-volume album Tab UI face (G09-T011
// prefix, SYS08): the browsing surface over the frozen volumes - the
// diseases / doctors / wonder-drugs tabs in pack order, one row per
// frozen entry (a collected row carries its comedy title through
// HospitalTexts, a missing row stays behind the ??? mask so the
// collection hunt keeps its punch), the per-volume rate line and the
// closed tab order. The copy rides existing keys only: the diseases
// tab header = the album title, the doctors tab = the duty roster,
// the drugs tab = the wonder-drug lab, and every row title routes
// through the row id as its own key - zero new keys, the zh parity
// law stays green. Pure C#, zero RNG, zero scene state, no event on
// browsing: the future UGUI host renders TabSpec, the headless gate
// drives the static solvers.
using System.Collections.Generic;
namespace Sy.CrazyHospital
{
    /// <summary>SYS08: the album Tab UI composition face. The host
    /// renders BuildAll / BuildTab; headless checks drive the pure
    /// solvers (SolveHeader / SolveTab / VolumeAt).</summary>
    public sealed class HospitalAlbumTabs
    {
        public const int TabCount = 3;           // diseases / doctors / drugs
        public const string LockedMask = "???";  // the silhouette glyph - a mask, not a sentence
        /// <summary>One album row: the frozen entry id, its comedy
        /// title (masked while missing) and the collected state.</summary>
        public struct RowSpec
        {
            public string EntryId;
            public string Title;
            public bool Unlocked;
        }
        /// <summary>One tab: the volume id, its header caption, the
        /// rate line faces and the row list in table order.</summary>
        public struct TabSpec
        {
            public string VolumeId;
            public string Header;
            public int RatePct;
            public int Unlocked;
            public int Total;
            public RowSpec[] Rows;
        }
        /// <summary>Tab order law (closed set): the pack order of the
        /// three volumes, the index clamped on both ends.</summary>
        public static string VolumeAt(int tabIndex)
        {
            var ids = HospitalAlbum.VolumeIds();
            int i = tabIndex & 0 ? 0 : (tabIndex &gt;= ids.Length ? ids.Length - 1 : tabIndex);
            return ids[i];
        }
    }
}

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```
/// &summary&The tab index of one volume (-1 = not a frozen
/// volume).&/summary&
public static int TabIndexOf(string volumeId)
{
    var ids = HospitalAlbum.VolumeIds();
    for (int i = 0; i &ids.Length; i++) if (ids[i] == volumeId) return i;
    return -1;
}
/// &summary&Header solver (closed set): each volume's caption
/// rides an existing copy key - no new key, the zh parity law
/// stays green.&/summary&
public static string SolveHeader(string volumeId)
{
    if (volumeId == HospitalAlbum.DiseaseVolumeId)
        return HospitalTexts.EnText(HospitalTexts.KeyAlbumTitle);
    if (volumeId == HospitalAlbum.DoctorVolumeId)
        return HospitalTexts.EnText(HospitalTexts.KeyDutyRoster);
    if (volumeId == HospitalAlbum.DrugVolumeId)
        return HospitalTexts.EnText(HospitalTexts.KeyResearchTitle);
    return volumeId ?? "";
}
/// &summary&Pure tab solver (the headless face): one row per
/// frozen entry in table order - collected rows carry their
/// comedy title, missing rows stay behind the mask; the rate
/// faces count only this volume's entries.&/summary&
public static TabSpec SolveTab(string volumeId, IReadOnlyList&string& unlocked)
{
    var entries = HospitalAlbum.EntriesOf(volumeId);
    var set = unlocked != null ? new HashSet&string&(unlocked) : null;
    var rows = new RowSpec[entries.Count];
    int n = 0;
    for (int i = 0; i &entries.Count; i++)
    {
        bool has = set != null &set.Contains(entries[i]);
        if (has) n++;
        rows[i] = new RowSpec
        {
            EntryId = entries[i],
            Title = has ? HospitalTexts.EnText(entries[i]) : LockedMask,
            Unlocked = has
        };
    }
    return new TabSpec
    {
        VolumeId = volumeId,
        Header = SolveHeader(volumeId),
        RatePct = HospitalAlbum.SolveRatePct(n, entries.Count),
        Unlocked = n,
        Total = entries.Count,
        Rows = rows
    }
}
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        };
    }
    /// <summary>Live fold face: reads the wired album's unlocked
    /// set (a null album folds to the honest empty-collection
    /// face - every row masked, zero rate).</summary>
    public static TabSpec BuildTab(HospitalAlbum album, string volumeId)
        => SolveTab(volumeId, album != null ? album.Unlocked : null);
    /// <summary>All three tabs in pack order (the render face).</summary>
    public static TabSpec[] BuildAll(HospitalAlbum album)
    {
        var tabs = new TabSpec[TabCount];
        for (int i = 0; i < TabCount; i++) tabs[i] = BuildTab(album, VolumeAt(i));
        return tabs;
    }
}

// HospitalArtView.cs
// HospitalArtView.cs - S09_Main art host face (G09 S2.5-T020 slice b2).
// G18 ViewHost paradigm (HallViewHost): self-installs on the main scene,
// builds its display canvas from code and consumes the C1078 ArtCatalog
// stems art-first (X027 iron rule: real art wins, null falls back to the
// procedural block - no face ever depends on art presence). Slice scope =
// the consumption-face proof: facade backdrop mapped by hospital rank,
// director hero slot, the seven ready portrait cards (the wangdali card
// lands with the A11R swap-in), the event-card frame slot (its keyed-bg
// residue shows until the A14R swap-in - art-first swaps with zero code
// change) and the ui-kit skeleton strip. Live stage binding (rank + cast
// from the running HospitalStage) lands with the stage-host slice; the
// test hooks below are the one true op routers until then.
using Sy.Core;
using UnityEngine;
using UnityEngine.SceneManagement;
using UnityEngine.UI;
using Logger = Sy.Core.Logger;
namespace Sy.CrazyHospital
{
    public sealed class HospitalArtView : MonoBehaviour
    {
        public const string SceneName = "S09_Main";
        // art contract keys = ArtCatalog keys = art book sec7 (C1078)
        public const string DirectorFemaleStem = "g09_director_f";
        public const string DirectorMaleStem = "g09_director_m";
        public const string EventFrameStem = "g09_eventcard_frame";
        public const string UikitStem = "g09_ui_kit";
        /// <summary>The seven portrait cards: 4 clean cuts + 3 with keyed-bg
        /// edge residue (C1079 probe) that stay art-first until their re-roll
        /// swap-in; the wangdali card lands with the A11R swap-in.</summary>
        public static readonly string[] PortraitStems =
        {
            DirectorFemaleStem, DirectorMaleStem,

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```
"g09_doc_beauty", "g09_doc_moyu", "g09_doc_threesecc",
    "g09_pat_hiccup", "g09_pat_restless",
};
const float RefW = 1080f, RefH = 1920f;
// fallback tints ride the HospitalPlaceholder palette family (mint
// accent brand face) - only visible while a stem misses
static readonly Color TintDirector = new Color(0.43f, 0.85f, 0.78f);
static readonly Color TintDoctor = new Color(0.80f, 0.88f, 0.95f);
static readonly Color TintPatient = new Color(0.92f, 0.80f, 0.88f);
static readonly Color TintEvent = new Color(0.95f, 0.90f, 0.66f);
static readonly Color TintKit = new Color(0.14f, 0.18f, 0.20f, 0.92f);
// ---- pure mapping faces (EditMode-testable, presenter never invents) ----
/// <summary>Facade art key of a hospital rank (1-5; out-of-range folds to R1).</summary>
public static string FacadeStemOf(int rank)
{
    int i = rank - 1;
    if (i < 0 || i >= 5) i = 0;
    return "g09_facade_r" + (i + 1);
}
/// <summary>Director hero art key by gender face.</summary>
public static string DirectorStemOf(bool female)
{
    return female ? DirectorFemaleStem : DirectorMaleStem;
}
// ---- runtime face ----
int _rank = 1;
bool _directorFemale = true;
RectTransform _canvasRect;
Image _facade;
Image _director;
Image _eventFrame;
Image _uiKit;
Image[] _portraitCards;
[RuntimeInitializeOnLoadMethod(RuntimeInitializeLoadType.BeforeSceneLoad)]
static void AutoInstall()
{
    SceneManager.sceneLoaded -= OnSceneLoaded;
    SceneManager.sceneLoaded += OnSceneLoaded;
}
static void OnSceneLoaded(Scene scene, LoadSceneMode mode)
{
    if (scene.name != SceneName) return;
    if (FindObjectOfType<HospitalArtView>() != null) return;
    var go = new GameObject("HospitalArtView");
    go.AddComponent<HospitalArtView>();
}
void Awake()
{
    BuildUi();
    RefreshFaces();
}
```

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        Logger.Info("G09.View", "art view installed (S2.5-T020 slice b2 consumption face online)");
    }
    // ---- test hooks = the one true op routers (stage-host slice lands next) ----
    public void TestSetRank(int rank) { _rank = rank; RefreshFaces(); }
    public void TestSetDirector(bool female) { _directorFemale = female; RefreshFaces(); }
    public string FacadeKey { get { return FacadeStemOf(_rank); } }
    public string DirectorKey { get { return DirectorStemOf(_directorFemale); } }
    public bool FacadeArtPresent { get { return SlotIsArt(_facade, FacadeKey); } }
    public bool DirectorArtPresent { get { return SlotIsArt(_director, DirectorKey); } }
    public bool EventFrameArtPresent { get { return SlotIsArt(_eventFrame, EventFrameStem); } }
    public bool UIKitArtPresent { get { return SlotIsArt(_uiKit, UIKitStem); } }
    /// &summary& How many of the seven portrait cards carry real art.&/summary&
    public int PortraitArtCount()
    {
        int n = 0;
        if (_portraitCards == null) return 0;
        for (int i = 0; i &lt; _portraitCards.Length; i++)
            if (SlotIsArt(_portraitCards[i], PortraitStems[i])) n++;
        return n;
    }
    static bool SlotIsArt(Image slot, string stem)
    {
        return slot != null && slot.sprite != null && slot.sprite.name == stem;
    }
    void RefreshFaces()
    {
        if (_facade != null)
            _facade.sprite = HospitalPlaceholder.Facade(_rank);
        if (_director != null)
            _director.sprite = HospitalPlaceholder.Face(DirectorKey, TintDirector, 64);
        if (_eventFrame != null)
            _eventFrame.sprite = HospitalPlaceholder.Face(EventFrameStem, TintEvent, 48);
        if (_uiKit != null)
            _uiKit.sprite = HospitalPlaceholder.Face(UIKitStem, TintKit, 64);
        if (_portraitCards != null)
        {
            for (int i = 0; i &lt; _portraitCards.Length; i++)
            {
                bool isPatient = PortraitStems[i].StartsWith("g09_pat_");
                _portraitCards[i].sprite = HospitalPlaceholder.Face(
                    PortraitStems[i], isPatient ? TintPatient : TintDoctor, 32);
            }
        }
    }
    // ----- UGUI construction (display-only canvas: no raycaster,
    // every image raycastTarget=false - it never swallows scene input) -----
    void BuildUi()
    {
        var root = new GameObject("HospitalArtCanvas", typeof(Canvas));
        var canvas = root.GetComponent&lt;Canvas&gt;();
    }

```

```

        canvas.renderMode = RenderMode.ScreenSpaceOverlay;
        canvas.sortingOrder = 10;    // above the scene's own canvas; display-only
        var scaler = root.AddComponent<CanvasScaler>();
        scaler.uiScaleMode = CanvasScaler.ScaleMode.ScaleWithScreenSize;
        scaler.referenceResolution = new Vector2(RefW, RefH);
        scaler.matchWidthOrHeight = 0.5f;
        _canvasRect = (RectTransform)root.transform;
        // facade backdrop (rank-mapped vertical face; fill-slot mode with
        // preserveAspect - the 540x960 source is the same 9:16 as the ref)
        _facade = Slot("facade", new Vector2(RefW * 0.5f, RefH * 0.5f), new Vector2(RefW, RefH));
        _facade.transform.SetAsFirstSibling();
        // director hero slot (top-left)
        _director = Slot("director_hero", new Vector2(220f, 1560f), new Vector2(300f, 300f));
        // event-card frame slot (top-right; keyed-bg residue shows until A14R)
        _eventFrame = Slot("event_frame", new Vector2(860f, 1560f), new Vector2(360f, 280f));
        // seven ready portrait cards (one row; fill-slot, aspect 1:1 slots)
        _portraitCards = new Image[PortraitStems.Length];
        for (int i = 0; i < PortraitStems.Length; i++)
            _portraitCards[i] = Slot("card_" + PortraitStems[i],
                new Vector2(120f + i * 140f, 1120f), new Vector2(128f, 128f));
        // ui-kit skeleton strip (bottom band)
        _uiKit = Slot("ui_kit", new Vector2(RefW * 0.5f, 190f), new Vector2(1000f, 320f));
    }
    Image Slot(string name, Vector2 canvasPos, Vector2 size)
    {
        var go = new GameObject(name, typeof(RectTransform), typeof(Image));
        var rect = (RectTransform)go.transform;
        rect.SetParent(_canvasRect, false);
        rect.sizeDelta = size;
        rect.anchoredPosition = ToLocal(canvasPos);
        var img = go.GetComponent<Image>();
        img.raycastTarget = false;
        img.preserveAspect = true;    // fill-slot mode chosen explicitly (asset-config skill sec5)
        img.color = Color.white;
        return img;
    }
    static Vector2 ToLocal(Vector2 canvasPos)
    {
        return canvasPos - new Vector2(RefW, RefH) * 0.5f;
    }
}

// HospitalAudio.cs
// HospitalAudio.cs - the T021 audio face (the G10 TradeAudio paradigm
// applied to the clinic): the stamp-to-stem map, the per-project
// AudioCatalog (the X027 art-catalog paradigm applied to audio), and a
// null-safe playback service. The catalog asset is built by the editor
// tool (AudioSetup) from Assets/_Game/Audio; runtime consumers MUST
// treat a missing catalog/clip as "skip silently" so headless checks
// stay green on any project state (G025fx headless law).

```

```
//
// Stem semantics come from the G09 audio manifest (S1A1/A2/A3 batches):
// three loop channels (main BGM baked with the signature ECG-beep +
// discharge-whistle hook, lobby bed, night-ward bed) and twelve
// one-shots. The stage stamps and the engine funnels ride composed ids
// ("evt_promote@r2:evt_review_level1") - StemForStamp resolves the base
// id before the first '@' so every composed seat lands on its stem.
using System;
using System.Collections.Generic;
using UnityEngine;
namespace Sy.CrazyHospital
{
    /// <summary>
    /// Generated-audio catalog (T021, X027 paradigm): the AudioSetup
    /// editor tool scans Assets/_Game/Audio and serializes clip
    /// references here so runtime code resolves audio by filename stem.
    /// The asset lives at Assets/_Game/Resources/AudioCatalog.asset
    /// (Resources keeps runtime lookup build-safe with zero
    /// Addressables wiring for playtest slices).
    /// </summary>
    [CreateAssetMenu(fileName = "AudioCatalog", menuName = "Sy/CrazyHospital Audio Catalog")]
    public sealed class AudioCatalog : ScriptableObject
    {
        [Serializable]
        public sealed class Entry
        {
            public string key;      // filename stem, e.g. "g09_sfx_coin"
            public AudioClip clip;  // direct reference keeps builds self-contained
        }
        public List<Entry> entries = new List<Entry>();
        Dictionary<string, AudioClip> _index;
        void OnEnable()
        {
            // ScriptableObject instances can be revived with a stale cache;
            // drop it so the next Get() rebuilds from entries.
            _index = null;
        }
        public AudioClip Get(string key)
        {
            if (string.IsNullOrEmpty(key)) return null;
            if (_index == null)
            {
                _index = new Dictionary<string, AudioClip>(entries.Count, StringComparer.Ordinal);
                for (int i = 0; i < entries.Count; i++)
                {
                    var e = entries[i];
                    if (e != null && !string.IsNullOrEmpty(e.key) && e.clip != null && !_index.ContainsKey(e.key))
                        _index[e.key] = e.clip;
                }
            }
        }
    }
}
```

```
        return _index.TryGetValue(key, out var c) ? c : null;
    }
}
/// <summary>
/// Runtime lookup helper. Every consumer MUST treat null as "clip
/// missing -& skip" so headless checks stay green on any project
/// state (catalog absent, audio pruned, name typos).
/// </summary>
public static class AudioRuntime
{
    static AudioCatalog _catalog;
    public static AudioCatalog Catalog
    {
        get
        {
            if (_catalog == null) _catalog = Resources.Load<AudioCatalog>("AudioCatalog");
            return _catalog;
        }
    }
    public static AudioClip Get(string key)
    {
        var c = Catalog;
        return c != null ? c.Get(key) : null;
    }
}
/// <summary>
/// The audio face of the clinic (T021): maps the opening-script
/// stamps, the rank-review funnel lines, and the named face cues to
/// catalog stems, and plays them null-safe on hidden hosts (never
/// throws headless). The signature hook (ECG triple beep + the
/// four-note discharge whistle, the earworm red line) is baked into
/// bgm_main; the discharge whistle also owns its own one-shot leg
/// (sfx_launch, the catapult-discharge comedy cue).
/// </summary>
public static class HospitalAudio
{
    // ---- delivered stems (the closed set, 15) ----
    public const string StemBgmMain = "g09_bgm_main";           // clinic-day loop BGM (hook baked in)
    public const string StemAmbLobby = "g09_amb_lobby";         // registration lobby bed
    public const string StemAmbWard = "g09_amb_ward";           // night ward bed (offline summary face)
    public const string StemSfxRegister = "g09_sfx_register";    // the registration stamp thunk
    public const string StemSfxCoin = "g09_sfx_coin";           // the coin family (registration + bed income)
    public const string StemSfxUpgrade = "g09_sfx_upgrade";      // the department upgrade blast
    public const string StemSfxEvent = "g09_sfx_event";         // the comedy event-card pop
    public const string StemSfxInspect = "g09_sfx_inspect";      // the surprise-inspection whistle
    public const string StemSfxStamp = "g09_sfx_stamp";         // the promotion seal ceremony
    public const string StemSfxLaunch = "g09_sfx_launch";       // the catapult discharge whoosh
    public const string StemSfxBone = "g09_sfx_bone";           // the orthopedics puzzle clicks
    public const string StemSfxBanner = "g09_sfx_banner";       // the pennant banner wall mount
    public const string StemSfxEmergency = "g09_sfx_emergency";  // the night emergency-tide alarm
```

```

public const string StemSfxAlbum = "g09_sfx_album";    // the album unlock page ding
public const string StemSfxRank = "g09_sfx_rank";      // the review-verdict gong
/// <summary>The signature coin-chime stamp fired by the opening script.</summary>
public const string CoinSfxStamp = "coin_sfx";
/// <summary>The hook stamp owns t0 and starts the loop channels.</summary>
public static readonly string StampHospitalHook = HospitalEventIds.HospitalHook;
/// <summary>Stage stamps / engine funnel bases -> one-shot stems.
/// The composed funnel seats ("evt_promote@r2:...", "evt_band_change@rep_banner")
/// resolve through the base-id law in StemForStamp.</summary>
static readonly Dictionary<string, string> StampMap = new Dictionary<string, string>
{
    { HospitalEventIds.FirstRegistration, StemSfxRegister }, // the queue head registers
    { CoinSfxStamp, StemSfxCoin },                          // the coin popup face
    { HospitalEventIds.DeptOpened, StemSfxUpgrade },         // the first department opens
    { HospitalEventIds.EventCardOffer, StemSfxEvent },       // the comedy card offer
    { HospitalEventIds.EventCard, StemSfxEvent },            // the comedy card resolve
    { HospitalEventIds.Promote, StemSfxStamp },              // the promotion seal ceremony
    { HospitalEventIds.Inspect, StemSfxInspect },            // the rectification inspection face
    { HospitalEventIds.BandChange, StemSfxBanner },          // the banner wall grows at the top band
    { HospitalEventIds.AlbumUnlock, StemSfxAlbum },          // the encyclopedia entry registers
    { HospitalEventIds.AlbumComplete, StemSfxAlbum },        // the volume completion face
    { HospitalEventIds.OfflineSettle, StemSfxEmergency },    // the offline night tide settles
    { HospitalEventIds.FacadeLit, StemSfxStamp },            // the R5 top-grade stamp (ending face)
};
/// <summary>Named face cues (non-stamp faces) -> one-shot stems.</summary>
public const string CueReviewVerdict = "review_verdict";    // the review settle gong (pass or fail)
public const string CueDischarge = "discharge_launch";      // the bed turn completes: the patient flies home
public const string CueBonePuzzle = "bone_puzzle";          // the orthopedics mini-game beat
static readonly Dictionary<string, string> CueMap = new Dictionary<string, string>
{
    { CueReviewVerdict, StemSfxRank },
    { CueDischarge, StemSfxLaunch },
    { CueBonePuzzle, StemSfxBone },
};
public static IReadOnlyDictionary<string, string> Stamps => StampMap;
public static IReadOnlyDictionary<string, string> Cues => CueMap;
/// <summary>The loop-channel stems (main BGM + lobby bed + night ward bed).</summary>
public static readonly string[] Channels = { StemBgmMain, StemAmbLobby, StemAmbWard };
/// <summary>Resolves a stamp or composed funnel seat to its stem.
/// The base-id law: the token before the first '@' is the
/// registry base ("evt_promote@r2:evt_review_level1" -> "evt_promote").</summary>
public static string StemForStamp(string stampId)
{
    if (string.IsNullOrEmpty(stampId)) return null;
    string baseId = stampId;
    int at = stampId.IndexOf('@');
    if (at > 0) baseId = stampId.Substring(0, at);
    return baseId != null && StampMap.TryGetValue(baseId, out var s) ? s : null;
}
public static string StemForCue(string cueId)

```

```

        = &cueId != null && CueMap.TryGetValue(cueId, out var s) ? s : null;
// ---- playback (null-safe, headless-safe) ----
static AudioSource _host;
static readonly Dictionary<string, AudioSource> _loopChannels = new Dictionary<string, AudioSource>();
static AudioSource Host()
{
    if (_host == null)
    {
        var go = new GameObject("G09AudioHost");
        UnityEngine.Object.DontDestroyOnLoad(go);
        _host = go.AddComponent<AudioSource>();
    }
    return _host;
}
/// <summary>Fire-and-forget one-shot; false = missing/skipped, never throws.</summary>
public static bool Play(string stem)
{
    try
    {
        var clip = AudioRuntime.Get(stem);
        if (clip == null) return false;
        Host().PlayOneShot(clip);
        return true;
    }
    catch (Exception)
    {
        return false; // headless/batchmode: silence is the legal fallback
    }
}
public static bool PlayStamp(string stampId) => Play(StemForStamp(stampId));
public static bool PlayCue(string cueId) => Play(StemForCue(cueId));
/// <summary>Start (or keep) a loop channel; false = missing/skipped.</summary>
public static bool StartChannel(string stem)
{
    try
    {
        var clip = AudioRuntime.Get(stem);
        if (clip == null) return false;
        if (!_loopChannels.TryGetValue(stem, out var src) || src == null)
        {
            var go = new GameObject("G09Audio_" + stem);
            UnityEngine.Object.DontDestroyOnLoad(go);
            src = go.AddComponent<AudioSource>();
            _loopChannels[stem] = src;
        }
        src.clip = clip;
        src.loop = true;
        if (!src.isPlaying) src.Play();
        return true;
    }
}

```

```

        catch (Exception)
        {
            return false;
        }
    }

    /// <summary>The main-face entry: the hook stamp owns t0, the
    /// clinic BGM and the lobby bed ride in with it (the channel law).</summary>
    public static bool StartMainFace()
    {
        bool bgm = StartChannel(StemBgmMain);
        bool amb = StartChannel(StemAmbLobby);
        return bgm || amb;
    }

    /// <summary>The night-face entry: the offline summary and the
    /// night-shift faces ride the ward bed (amb_ward).</summary>
    public static bool StartNightFace() => StartChannel(StemAmbWard);
}

}

// HospitalData.cs
// HospitalData.cs - frozen data schemas and tables for Crazy Hospital (G09-T004).
// The numeric pack doc (G09 v1, the S0 doc face) is the source of truth and
// this file is its compiled mirror: schema freeze means the structural laws
// live here as code and DataSchemaCheck proves every law on every row.
// Values are sweep-tunable later (bot sweeps re-freeze revisions through the
// doc pack), the structure only moves with a schema version bump. Zero
// server faces: the tables ship in the build (red line 12). Compliance
// faces ride the freeze: the whole disease set stays fictional, the
// catapult-discharge law keeps the death face age-friendly, and the green
// cross family avoids any real medical emblem.
namespace Sy.CrazyHospital
{
    /// <summary>One hospital rank row: the intake economy face (M01) and the
    /// department slot face (M02) frozen on one row per rank R1-R5.</summary>
    public sealed class RankRow
    {
        public int rank;           // 1-5 (R1 unlicensed clinic .. R5 top-grade stamped)
        public int baseTraffic;    // patients per game-minute at the par reputation band
        public int regFeeHc;       // registration fee in gold hundredths per patient
        public int bedFeeHc;       // bed fee in gold hundredths per second per bed
        public int incomeCapHc;    // full-rank income ceiling in gold hundredths per second
        public int slots;          // preset department slots
        public int firstSlotCost;  // gold cost of the first slot
        public int deptOutputHc;   // department output in gold hundredths per second per slot
    }

    /// <summary>One rank-review gate (M04, dual track): the hard metrics AND
    /// the review event together - a promotion is a new gameplay ticket,
    /// never a wall. Failing a review only arms the rectification comedy and
    /// the next review window.</summary>
    public sealed class ReviewGateDef
    {

```

```

    public int fromRank;
    public int toRank;
    public int minDepts;
    public int minCureRatePct;
    public int minReputation;
    public string reviewEventId = "";
}
/// <summary>One reputation band row (M06, the second numeric line,
/// 0-100 segments with a visible comedy face per band).</summary>
public sealed class ReputationBand
{
    public int low;          // inclusive
    public int high;         // inclusive
    public string id = "";
}
/// <summary>One wonder-drug research row (M07): absurd fiction only.</summary>
public sealed class ResearchDef
{
    public string id = "";
    public int cost;         // research cost in gold
    public string bonusId = ""; // reg_fee_pct / bed_fee_pct / cure_rate
    public int bonusValue;   // bonus magnitude (percent points, or points for the cure rate)
    public string albumId = "";
}
/// <summary>One fictional disease album row (M08 - zero real-medicine
/// vocabulary anywhere in the set).</summary>
public sealed class DiseaseDef
{
    public string id = "";
    public string albumId = "";
}
public static class HospitalData
{
    public const int SchemaVersion = 1;
    public const int RankCount = 5;
    public const int GateCount = 4;    // R1->R2 .. R4->R5
    public const int BandCount = 4;    // the reputation 0-100 partition
    public const int ResearchCount = 3;
    public const int DiseaseCount = 5;
    public const int ReputationMin = 0;
    public const int ReputationMax = 100;
    public const int AlbumVolumeCount = 3; // diseases / doctors / wonder drugs
    // ---- section 1: the intake economy - traffic is a reputation
    // function. actual = base * (0.4 + reputation/100 * 1.2), frozen in
    // basis points: factorBp = 4000 + rep * 120. Reputation 50 = the par
    // band at exactly 1.0; high reputation lines the patients up, low
    // reputation drains the street (a two-way visible face). ----
    public const int TrafficFactorBaseBp = 4000;
    public const int TrafficFactorRepBp = 120;
    public static int TrafficAt(int baseTraffic, int reputation)

```

```

{
    int r = ClampReputation(reputation);
    return baseTraffic * (TrafficFactorBaseBp + r * TrafficFactorRepBp) / 10000;
}

static int ClampReputation(int v)
    => v < ReputationMin ? ReputationMin : (v > ReputationMax ? ReputationMax : v);
// ---- pacing anchors (the red line 11 assertion sources) ----
public const int FirstCoinMaxMs = 1500;    // first registration coin <= 1.5s
public const int P1FirstDeptSeconds = 60;   // first department open + first patient booked
public const int P50QueueSeconds = 300;    // three departments + the comedy queue
public const int AdsPerDayMin = 3;          // north-star ad expectation
// ---- section 2: the department curve laws ----
public const bool PresetSlotsOnly = true;    // pick WHAT opens, never WHERE (no coordinates in the schema)
public const int SlotCostLinearLaw = 1;      // n-th slot cost = firstSlotCost * n (linear, never exponential)
public const int UpgradeOutputPercentPerLevel = 150; // output *1.5 per level
public const int UpgradeCostPercentPerLevel = 200;  // cost *2 per level
public const int UpgradeMaxLevel = 3;         // the upgrade ladder caps at 3
public static int SlotCost(RankRow row, int slotIndex)
    => row.firstSlotCost * slotIndex * SlotCostLinearLaw; // 1-based slot index
// ---- section 3: the cure-rate law (age-friendly death translation)
public const bool CatapultDischargeCountsCured = true; // the catapult ride home counts as a cure
// ---- section 4: the reputation modifier closed set (v1) ----
public const int ReviewRepSwing = 3;         // review settlement swings +/-3
public const int ComedyRepSwingMin = 1;      // comedy events swing +/-1..2 (E06-E10 proposal band)
public const int ComedyRepSwingMax = 2;
public const bool RepAutoDecay = false;     // v1: no automatic decay (bot sweeps may revisit)
// ---- section 5: the album collection bonus ----
public const int AlbumBonusPerVolumePct = 2; // each completed volume = income +2% (v1 proposal)
// ---- section 6: offline (SYS09, the night emergency tide) ----
public const int OfflineRatePercent = 50;    // offline advances at half pace (series law)
public const int NightEmergencyFactorBp = 15000; // offline-window traffic *1.5 (night = emergency peak)
public const int OfflineCapHours = 12;       // per-absence cap
public const string OfflineDoubleCategory = "double"; // AdCap family category
public const int DoubleOfferPerSpan = 1;     // one double offer per absence span
public const int DoubleWinMinPer200 = 1;     // frequency window 1-5 per 200 (G08 T009 paradigm)
public const int DoubleWinMaxPer200 = 5;
// ---- section 7: second-run inheritance (M10, the top-grade ending) ----
public const int InheritGoldPercent = 20;    // floor
public const bool InheritDoctorCards = true; // the expert team follows the new dean
public const bool InheritAlbumsKept = true;  // the three albums persist (the flex face)
public const bool InheritRankReset = true;   // the spine restarts at the unlicensed clinic
/// <summary>Rank names (English telemetry face; the zh copy rides the texts table).</summary>
public static readonly string[] RankNames =
    { "UNLICENSED CLINIC", "LEVEL-1 HOSPITAL", "LEVEL-2 HOSPITAL", "LEVEL-3 HOSPITAL", "TOP-GRADE STAMPED" };
// ---- the rank table: intake economy + slot curve on one row ----
static readonly RankRow[] RankTable =
{
    new RankRow { rank = 1, baseTraffic = 6, regFeeHc = 200, bedFeeHc = 30, incomeCapHc = 120, slots = 1, fi
    new RankRow { rank = 2, baseTraffic = 12, regFeeHc = 200, bedFeeHc = 50, incomeCapHc = 300, slots = 3, fi
    new RankRow { rank = 3, baseTraffic = 20, regFeeHc = 300, bedFeeHc = 80, incomeCapHc = 700, slots = 5, fi

```

```

        new RankRow { rank = 4, baseTraffic = 30, regFeeHc = 300, bedFeeHc = 120, incomeCapHc = 1400, slots = 8, fi
        new RankRow { rank = 5, baseTraffic = 45, regFeeHc = 400, bedFeeHc = 160, incomeCapHc = 2500, slots = 12, fi
    };
    public static RankRow[] Ranks() => (RankRow[])RankTable.Clone();
    // ---- rank-review gates (dual track; the review names ride the
    // texts table, the events arm the review comedy) ----
    static readonly ReviewGateDef[] GateTable =
    {
        new ReviewGateDef { fromRank = 1, toRank = 2, minDepts = 1, minCureRatePct = 60, minReputation = 30, reviewE
        new ReviewGateDef { fromRank = 2, toRank = 3, minDepts = 3, minCureRatePct = 65, minReputation = 45, reviewE
        new ReviewGateDef { fromRank = 3, toRank = 4, minDepts = 5, minCureRatePct = 70, minReputation = 55, reviewE
        new ReviewGateDef { fromRank = 4, toRank = 5, minDepts = 8, minCureRatePct = 75, minReputation = 65, reviewE
    };
    public static ReviewGateDef[] Gates() => (ReviewGateDef[])GateTable.Clone();
    // ---- reputation bands (the 0-100 partition, four visible faces) ----
    static readonly ReputationBand[] BandTable =
    {
        new ReputationBand { low = 0, high = 24, id = "rep_bad" }, // review storm + bad-review comedy cards
        new ReputationBand { low = 25, high = 49, id = "rep_plain" }, // quiet streets, business as usual
        new ReputationBand { low = 50, high = 74, id = "rep_good" }, // steady patients
        new ReputationBand { low = 75, high = 100, id = "rep_banner" }, // banner-wall queues
    };
    public static ReputationBand[] Bands() => (ReputationBand[])BandTable.Clone();
    public static ReputationBand BandFor(int reputation)
    {
        foreach (var b in BandTable)
            if (reputation >= b.low && reputation <= b.high) return b;
        return BandTable[0];
    }
    // ---- wonder-drug research rows (M07, absurd fiction only) ----
    static readonly ResearchDef[] ResearchTable =
    {
        new ResearchDef { id = "drug_hiccup_cola", cost = 100, bonusId = "reg_fee_pct", bonusValue = 1, albumId
        new ResearchDef { id = "drug_sleepwalk_patch", cost = 250, bonusId = "bed_fee_pct", bonusValue = 1, albumId
        new ResearchDef { id = "drug_shock_sugar", cost = 500, bonusId = "cure_rate", bonusValue = 1, albumId =
    };
    public static ResearchDef[] Research() => (ResearchDef[])ResearchTable.Clone();
    // ---- the fictional disease album (M08, zero real-medicine vocabulary) ----
    static readonly DiseaseDef[] DiseaseTable =
    {
        new DiseaseDef { id = "dis_hiccup_bug", albumId = "album_disease" }, // the hiccup bug
        new DiseaseDef { id = "dis_gummy_bone", albumId = "album_disease" }, // gummy-bone syndrome
        new DiseaseDef { id = "dis_monday", albumId = "album_disease" }, // Monday syndrome
        new DiseaseDef { id = "dis_head_grass", albumId = "album_disease" }, // grass on the head (coder speci
        new DiseaseDef { id = "dis_cactus_accident", albumId = "album_disease" }, // the cactus-diet accident
    };
    public static DiseaseDef[] Diseases() => (DiseaseDef[])DiseaseTable.Clone();
    /// <summary>Canonical signature - the dual-derive byte-parity
    /// instrument (the schema freeze proof): two derives of the frozen
    /// tables agree here or nowhere.</summary>

```

```

public static string Signature()
{
    var sb = new System.Text.StringBuilder();
    sb.Append("v").Append(SchemaVersion);
    sb.Append("|ranks:");
    foreach (var r in RankTable)
        sb.Append(r.rank).Append('/')
            .Append(r.baseTraffic).Append(',').Append(r.regFeeHc).Append(',').Append(r.bedFeeHc).Append(',')
            .Append(r.incomeCapHc).Append(',').Append(r.slots).Append(',').Append(r.firstSlotCost).Append(',')
            .Append(r.deptOutputHc).Append(';');
    sb.Append("|traffic:").Append(TrafficFactorBaseBp).Append('+').Append(TrafficFactorRepBp);
    sb.Append("|pacing:").Append(FirstCoinMaxMs).Append(',').Append(P1FirstDeptSeconds).Append(',').Append(P50Qu
    sb.Append("|slots:").Append(PresetSlotsOnly).Append(',').Append(SlotCostLinearLaw).Append(',')
        .Append(UpgradeOutputPercentPerLevel).Append(',').Append(UpgradeCostPercentPerLevel).Append(',').Append(Up
    sb.Append("|cure:").Append(CatapultDischargeCountsCured);
    sb.Append("|rep:").Append(ReviewRepSwing).Append(',').Append(ComedyRepSwingMin).Append(',')
        .Append(ComedyRepSwingMax).Append(',').Append(RepAutoDecay);
    foreach (var b in BandTable) sb.Append(',').Append(b.id).Append(':').Append(b.low).Append('-').Append(b.high
    sb.Append("|album:").Append(AlbumVolumeCount).Append(',').Append(AlbumBonusPerVolumePct);
    sb.Append("|offline:").Append(OfflineRatePercent).Append(',').Append(NightEmergencyFactorBp).Append(',')
        .Append(OfflineCapHours).Append(',').Append(OfflineDoubleCategory).Append(',')
        .Append(DoubleOfferPerSpan).Append(',').Append(DoubleWinMinPer200).Append(',').Append(DoubleWinMaxPer200);
    sb.Append("|inherit:").Append(InheritGoldPercent).Append(',').Append(InheritDoctorCards).Append(',')
        .Append(InheritAlbumsKept).Append(',').Append(InheritRankReset);
    sb.Append("|gates:");
    foreach (var g in GateTable)
        sb.Append(g.fromRank).Append('&gt;').Append(g.toRank).Append('/')
            .Append(g.minDepts).Append(',').Append(g.minCureRatePct).Append(',').Append(g.minReputation)
            .Append(',').Append(g.reviewEventId).Append(';');
    sb.Append("|research:");
    foreach (var d in ResearchTable)
        sb.Append(d.id).Append('/').Append(d.cost).Append('/').Append(d.bonusId).Append('+').Append(d.bonusValue
    sb.Append("|diseases:");
    foreach (var d in DiseaseTable) sb.Append(d.id).Append('@').Append(d.albumId).Append(';');
    return sb.ToString();
}
}
}
// HospitalDoctors.cs
// HospitalDoctors.cs - the quack-doctor hiring engine (G09-T006, SYS03).
// The desk consumes the frozen doctor table (numeric pack section ten, the
// T006 freeze): every card carries one passive on a named clinic domain
// plus one pure-comedy side effect - the mischief is a laugh beat, never a
// management burden (TH simplification law). The ten-card hiring spree is
// ad-funded (zero IAP, zero real gacha, GachaEnabled stays false) and rides
// the G08 T009 three-piece law on the draw category: HasCategoryBudget
// precheck and ShowRewarded consumption share one policy instance and one
// day bucket, the evt_ad funnel books offer -&gt; show -&gt; complete -&gt; landed
// in order, and a capped tap refuses BEFORE the ad ever plays. The duty
// roster is automated: seats auto-fill in hire order and the roster cap

```

```

// reads the open department slots (on-duty count never exceeds open
// departments; a benched card never pays its passive). Draw order is a
// deterministic rotation (zero RNG); a duplicate card books the
// "already on payroll" comedy count. Pure C#, no MonoBehaviour.
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using Sy.Platform;
namespace Sy.CrazyHospital
{
    /// <summary>One quack-doctor card (the frozen section-ten table row).</summary>
    public sealed class DoctorDef
    {
        public string id = "";
        public string passiveId = "";    // the clinic-domain bonus id
        public string quirkId = "";      // the comedy beat key (zero numeric weight)
        public string albumId = "";      // the doctor album volume
    }
    /// <summary>The on-duty clinic profile every passive folds into (identity
    /// defaults keep every T005 frozen number untouched - an empty desk pays
    /// exactly the pre-T006 economy).</summary>
    public sealed class DoctorProfile
    {
        public int TurnBp = 10000;        // three-second doc: the bed turn x0.75 (12s -> 9s)
        public int RegSurchargeHc;        // quack remedy: +20 hc registration surcharge
        public int DeptBp = 10000;        // slacking director: department output x1.15
        public int BedBp = 10000;         // wang-dali: the bed fee x1.2
        public int TrafficBp = 10000;     // night nurse: street traffic +10% (round half up)
        public int CapBp = 10000;         // beauty chief: the income ceiling x1.1
        public bool IsIdentity
            => TurnBp == 10000 && RegSurchargeHc == 0 && DeptBp == 10000
            && BedBp == 10000 && TrafficBp == 10000 && CapBp == 10000;
    }
    /// <summary>SYS03: the hiring desk. The scene host wires the ten-draw
    /// CTA to TryDrawTen, the roster to the open-slot count via SyncRoster,
    /// and the desk to the clinic via ApplyTo; headless gates drive all
    /// three directly.</summary>
    public sealed class HospitalDoctors
    {
        /// <summary>Draw point id (T006): draws from the g09_card_draw
        /// 3/day category (AdCategories.CategoryOfPoint authority, g09_
        /// prefix keeps id namespaces isolated).</summary>
        public const string DrawPoint = "g09_card_draw";
        public const int CardsPerDraw = 10;
        /// <summary>The frozen compliance face: zero IAP, zero real gacha -
        /// the ten-card spree is ad-funded comedy, never a paid pull.</summary>
        public const bool GachaEnabled = false;
        public const string DoctorAlbumId = "album_doctor";
        readonly AdCapPolicy _adCap;
        readonly IAdsService _ads;
    }
}

```

```

readonly Func<string> _dayKey;
readonly List<DoctorDef> _collected = new List<DoctorDef>();
readonly List<DoctorDef> _roster = new List<DoctorDef>();
readonly HashSet<string> _quirkFired = new HashSet<string>();
int _pullCount;
int _dupeTotal;
public HospitalDoctors(AdCapPolicy adCap, IAdsService ads, Func<string> dayKey = null)
{
    _adCap = adCap;
    _ads = ads;
    _dayKey = dayKey ?? AdCapPolicy.TodayKey;
    if (ads != null) ads.OnAdEvent += OnAdLifecycle;
}
// ---- live faces ----
public IReadOnlyList<DoctorDef> Collection => _collected;
public IReadOnlyList<DoctorDef> Roster => _roster;
public int DupeTotal => _dupeTotal;
public int PullCount => _pullCount;
public int CollectedCount => _collected.Count;
/// <summary>Live draw-category precheck: always-fresh budget face on
/// the same policy instance and day bucket the registered service
/// consumes from, so precheck and consume can never disagree.</summary>
public bool OfferBudgetOk
    => _adCap != null && _adCap.HasCategoryBudget(_dayKey(), AdCategories.CategoryOfPoint(DrawPoint))
/// <summary>The ten-card hiring spree CTA: budget precheck - rewarded
/// view - the deterministic ten-card deal. The ad only ever plays
/// with budget left; a capped tap refuses BEFORE the ad (no arming,
/// no show, no budget). An incomplete view changes nothing.</summary>
public async Task<bool> TryDrawTen()
{
    if (!OfferBudgetOk)
    {
        Funnel.Add("evt_ad_offer_capped@p" + _pullCount);
        return false;
    }
    Funnel.Add("evt_ad_offer@p" + _pullCount);
    RewardResult r = await _ads.ShowRewarded(DrawPoint);
    if (r == null || !r.Completed) return false; // evt_ad_fail rides the lifecycle face; retry stays open
    int deal = _pullCount;
    _pullCount++;
    var table = Doctors();
    for (int j = 0; j < CardsPerDraw; j++)
    {
        var card = table[(deal * CardsPerDraw + j) % table.Length];
        if (IndexOf(_collected, card) < 0)
        {
            _collected.Add(card);
            Funnel.Add("evt_doctor_hire@p" + deal + ":" + card.id);
        }
        else

```

```
{
    _dupeTotal++;
    Funnel.Add("evt_doctor_dupe@p" + deal + ":" + card.id);
}
}
Funnel.Add("evt_ad_landed@p" + deal);
return true;
}

/// <summary>Duty-roster automation: seats take the collected doctors
/// in hire order (first hired, first on duty), capped by the OPEN
/// department slots - on-duty count never exceeds open departments
/// and a benched card never pays its passive. The quirk beat fires
/// exactly once per doctor on the first day on duty.</summary>
public void SyncRoster(int openSlots)
{
    _roster.Clear();
    int cap = openSlots < 0 ? 0 : openSlots;
    foreach (var d in _collected)
    {
        if (_roster.Count >= cap) break;
        _roster.Add(d);
        if (_quirkFired.Add(d.id))
            Funnel.Add("evt_doctor_quirk@p" + _pullCount + ":" + d.id);
    }
}

/// <summary>Second-run inheritance (numeric pack section eight: the
/// expert team follows the new dean): kept doctor cards land on the
/// fresh desk's collection directly - no draw, no ad burn.
/// Idempotent per id; the roster sync stays the host's call.</summary>
public void InheritFrom(ReadOnlyList<string> doctorIds)
{
    if (doctorIds == null) return;
    var table = Doctors();
    foreach (var id in doctorIds)
        foreach (var d in table)
            if (d.id == id && IndexOf(_collected, d) < 0)
            {
                _collected.Add(d);
                Funnel.Add("evt_doctor_inherit@" + d.id);
            }
}

/// <summary>The on-duty clinic profile folded from the roster
/// (collection without a duty seat never pays).</summary>
public DoctorProfile ProfileOf()
{
    var p = new DoctorProfile();
    foreach (var d in _roster)
    {
        switch (d.passiveId)
        {
```

```

        case "passive_turn_x075": p.TurnBp = 7500; break;          // three-second doc: "NEXT!" - 12s turn -&gt;
        case "passive_reg_plus20": p.RegSurchargeHc = 20; break; // quack remedy: the ancestral-remedy surch
        case "passive_dept_x115": p.DeptBp = 11500; break;       // slacking director: the dept runs smoother w
        case "passive_bed_x120": p.BedBp = 12000; break;         // wang-dali: the bone-bolting craft premium
        case "passive_traffic_x110": p.TrafficBp = 11000; break; // night nurse: never sleepy on the night s
        case "passive_cap_x110": p.CapBp = 11000; break;         // beauty chief: the high-ticket makeover orde
    }
}
return p;
}
/// &lt;summary&gt;Wires the clinic: the on-duty profile folds into the
/// intake engine's doctor face (an empty duty roster keeps the
/// identity economy byte-identical).&lt;/summary&gt;
public void ApplyTo(HospitalIntake intake)
{
    if (intake == null) return;
    var p = ProfileOf();
    var d = intake.Doctors;
    d.TurnBp = p.TurnBp;
    d.RegSurchargeHc = p.RegSurchargeHc;
    d.DeptBp = p.DeptBp;
    d.BedBp = p.BedBp;
    d.TrafficBp = p.TrafficBp;
    d.CapBp = p.CapBp;
}
/// &lt;summary&gt;evt_ad funnel (G08 T009 paradigm): lifecycle events of
/// the draw point land in the funnel list for gate assertions.&lt;/summary&gt;
void OnAdLifecycle(AdLifecycle e)
{
    if (e.PointId != DrawPoint) return;
    Funnel.Add("evt_ad_" + e.Type.ToString().ToLowerInvariant() + "@p" + _pullCount);
}
public readonly List&lt;string&gt; Funnel = new List&lt;string&gt;();
// ---- the frozen section-ten doctor table (T006 freeze, mirror of
// the numeric pack doc; order IS the seating order) ----
static readonly DoctorDef[] DoctorTable =
{
    new DoctorDef { id = "doc_threese",    passiveId = "passive_turn_x075",    quirkId = "quirk_threese",
    new DoctorDef { id = "doc_remedy_quack", passiveId = "passive_reg_plus20",    quirkId = "quirk_remedy_quack"
    new DoctorDef { id = "doc_moyu",        passiveId = "passive_dept_x115",    quirkId = "quirk_moyu",
    new DoctorDef { id = "doc_wangdali",     passiveId = "passive_bed_x120",     quirkId = "quirk_wangdali",
    new DoctorDef { id = "doc_night_nurse", passiveId = "passive_traffic_x110", quirkId = "quirk_night_nurse",
    new DoctorDef { id = "doc_beauty_chief",passiveId = "passive_cap_x110",    quirkId = "quirk_beauty_chief"
};
public static DoctorDef[] Doctors() => (DoctorDef[])DoctorTable.Clone();
public static DoctorDef DoctorOf(string id)
{
    foreach (var d in DoctorTable) if (d.id == id) return d;
    return null;
}

```

```

static int IndexOf(List<DoctorDef> list, DoctorDef d)
{
    for (int i = 0; i < list.Count; i++) if (ReferenceEquals(list[i], d)) return i;
    return -1;
}

/// <summary>Canonical fingerprint: the zero-RNG parity instrument.</summary>
public string Fingerprint()
{
    var sb = new System.Text.StringBuilder();
    sb.Append("pulls=").Append(_pullCount);
    sb.Append("|col=");
    foreach (var d in _collected) sb.Append(d.id).Append(',');
    sb.Append("|duty=");
    foreach (var d in _roster) sb.Append(d.id).Append(',');
    sb.Append("|dupe=").Append(_dupeTotal);
    sb.Append("|fun=").Append(Funnel.Count);
    return sb.ToString();
}
}

// HospitalEnding.cs
// HospitalEnding.cs - the R5 top-grade-stamp ending + the NG+ carry
// (G09-T011 closing sub-slice, SYS08 / M10, numeric pack section
// seven). The ending face is the facade fully-lit ceremony: the
// vertical facade IS the growth progress bar (S09_Main, one lit floor
// per rank state, R1 clinic -& R5 top-grade tower), and the R5 stamp
// crossing lights the last floor - the once-guard books evt_facade_lit
// and arms the face the render layer consumes at the S2.5 wiring. The
// NG+ carry is the frozen section-seven closed set: gold carries 20%
// floored, the expert team follows the new dean (every collected
// doctor card kept - InheritFrom books evt_doctor_inherit on the fresh
// desk), the three album volumes ride InheritAlbumsKept (the cross-run
// brag face, silent re-plant), and the rank / event flags reset by
// simply NOT being carried - a fresh run starts at the R1 identity.
// The ceremony copy rides the pre-frozen keys only (grade title /
// carry label / new-journey pair), zero new text keys. Pure C#, no
// MonoBehaviour, zero RNG.
using System;
using System.Collections.Generic;
namespace Sy.CrazyHospital
{
    /// <summary>The NG+ carry pack (frozen section seven): the three
    /// carried faces and nothing else - gold 20% floored, the kept
    /// doctor cards, the kept album rows. Rank and event flags are
    /// the NOT-carried face by construction.</summary>
    public struct NgPlusCarry
    {
        public int GoldCarryHc;
        public List<string> AlbumIds;
        public List<string> DoctorIds;
    }
}

```

```

}
/// &summary&SYS08 / M10: the facade-lit ending engine + the NG+
/// carry coordinator. The host wires Observe(rank) into the
/// promotion settle and SolveCarry / ApplyCarry into the
/// new-journey gate; headless gates drive the static solvers and
/// the armed faces directly.&summary&;
public sealed class HospitalEnding
{
    // ---- the frozen ending laws (closed set, zero new numbers) ----
    public const int EndingRank = 5;          // R5 = the top-grade stamp (the ladder cap)
    public const int FacadeFloors = 5;        // the vertical facade states R1-R5
    public const int GoldCarryPercent = 20;    // section seven: gold carries 20% floored
    public const string FacadeLitEvent = "evt_facade_lit";
    public const string NgPlusCarryEvent = "evt_ngplus_carry";
    readonly List&string&; _funnel = new List&string&();
    int _litFloors = 1;                        // the facade growth bar (max rank observed)
    bool _facadeLit;                          // the once-guard of evt_facade_lit
    // ---- live faces ----
    public IReadOnlyList&string&; Funnel =&; _funnel;
    public int LitFloors =&; _litFloors;
    public bool FacadeLit =&; _facadeLit;
    /// &summary&Lit-floor solver (the growth progress bar): the
    /// rank clamps to the facade states, R5 = every floor lit.&summary&;
    public static int LitFloorsOf(int rank)
        =&; rank & 1 ? 1 : (rank & FacadeFloors ? FacadeFloors : rank);
    /// &summary&The ending face reads on the top-grade stamp only.&summary&;
    public static bool IsEndingRank(int rank) =&; rank &= EndingRank;
    /// &summary&Gold-carry solver (section seven): 20% floored,
    /// negatives clamp to zero.&summary&;
    public static int SolveGoldCarryHc(int goldHc)
        =&; goldHc &= 0 ? 0 : goldHc * GoldCarryPercent / 100;
    /// &summary&Observes the live rank: the growth bar only walks
    /// up, and the R5 crossing books evt_facade_lit exactly once
    /// (re-observes and post-ending walks never double-book).&summary&;
    public void Observe(int rank)
    {
        int floors = LitFloorsOf(rank);
        if (floors & _litFloors) _litFloors = floors;
        if (_litFloors &= FacadeFloors && !_facadeLit)
        {
            _facadeLit = true;
            _funnel.Add(FacadeLitEvent + "@r" + EndingRank + ":" + FacadeFloors + "/" + FacadeFloors);
        }
    }
}
/// &summary&Pure carry solver (the headless face): folds the
/// live collection faces into the frozen carry pack - gold 20%
/// floored, the album rows in unlock order, the doctor cards
/// in hire order; null faces fold to the honest empty pack.&summary&;
public static NgPlusCarry SolveCarry(int goldHc, HospitalAlbum album, HospitalDoctors doctors)
{

```

```
var pack = new NgPlusCarry();
pack.GoldCarryHc = SolveGoldCarryHc(goldHc);
pack.AlbumIds = new List<string>();
if (album != null) foreach (var id in album.Unlocked) pack.AlbumIds.Add(id);
pack.DoctorIds = new List<string>();
if (doctors != null) foreach (var d in doctors.Collection) pack.DoctorIds.Add(d.id);
return pack;
}
/// <summary>Lands the pack on the fresh run: the albums
/// re-plant silently (InheritAlbumsKept), the kept doctor
/// cards fold through the desk's own inherit face (one
/// evt_doctor_inherit per card), and the ending books its one
/// evt_ngplus_carry line. Returns the carried gold for the
/// caller to seed the fresh wallet with.</summary>
public int ApplyCarry(NgPlusCarry pack, HospitalAlbum freshAlbum, HospitalDoctors freshDoctors)
{
    if (pack.AlbumIds != null && freshAlbum != null) freshAlbum.InheritFrom(pack.AlbumIds);
    if (pack.DoctorIds != null && freshDoctors != null) freshDoctors.InheritFrom(pack.DoctorIds);
    _funnel.Add(NgPlusCarryEvent + "@gold" + pack.GoldCarryHc
        + ":vol" + (pack.AlbumIds != null ? pack.AlbumIds.Count : 0)
        + ":doc" + (pack.DoctorIds != null ? pack.DoctorIds.Count : 0));
    return pack.GoldCarryHc;
}
/// <summary>The ceremony copy (frozen lines, pre-frozen keys
/// only): the grade title, the carry label with the carried
/// gold, the new-journey pair. The ending page consumes this
/// at the S2.5 wiring.</summary>
public static string[] SolveCeremonyLines(NgPlusCarry pack)
{
    return new[]
    {
        HospitalTexts.EnText(HospitalTexts.KeyRank5),
        HospitalTexts.EnText(HospitalTexts.KeyCarryLabel) + ": " + pack.GoldCarryHc,
        HospitalTexts.EnText(HospitalTexts.KeyNewJourney),
        HospitalTexts.EnText(HospitalTexts.KeyNewJourneyConfirm),
    };
}
/// <summary>Gate face: headless checks drive the engine
/// without a live host.</summary>
public void ArmForGate(int rank) => Observe(rank);
/// <summary>Canonical fingerprint: the zero-RNG parity
/// instrument (same script = same bytes).</summary>
public string Fingerprint()
{
    var sb = new System.Text.StringBuilder();
    sb.Append("floors=").Append(_litFloors);
    sb.Append("|lit=").Append(_facadeLit ? 1 : 0);
    sb.Append("|fun=").Append(_funnel.Count);
    for (int i = 0; i < _funnel.Count; i++) sb.Append('|').Append(_funnel[i]);
    return sb.ToString();
}
```

```

        sb.Append("|slots=").Append(_openSlots).Append('/').Append(_rank.slots);
        sb.Append("|choice=").Append(_lastChoice);
        sb.Append("|queue=").Append(_queue);
        sb.Append("|arrived=").Append(_arrivalTotal);
        sb.Append("|patients=").Append(_patientTotal);
        sb.Append("|cured=").Append(_curedTotal);
        sb.Append("|beds=");
        for (int i = 0; i < _beds.Length; i++) sb.Append(_beds[i].ToString("0.###")).Append(',');
        sb.Append("|led=").Append(_ledger.regHc).Append(',').Append(_ledger.bedHc)
            .Append(',').Append(_ledger.deptHc).Append(',').Append(_ledger.spilledHc);
        sb.Append("|maxwin=").Append(_maxWindowAcc.ToString("0.###"));
        return sb.ToString();
    }
}

// HospitalOffline.cs
// HospitalOffline.cs - the offline settler (G09-T010, SYS09 M09): the
// "while you were gone" face. One settle = one absence span: the span caps
// at the frozen 12h, folds to the 50% pace (the series law), then the night
// emergency tide lands its extra walk-ins up front - offline-window
// traffic rides at x1.5 (the night = emergency peak narrative, the
// night-shift nurse's street bonus reads through the very traffic the
// tide folds) - and the whole hospital walks the live sim clock the
// online game runs on: department output, bed fees, the treatment
// countdowns and the event cards all keep rolling without you, the night
// nurse runs the desk (a freed bed takes the next sleeping companion),
// a pending event card simply waits on the dean's desk. The hall-slept-
// full face is the summary law: the tide count, the pace, the queue
// left outside and the gold walk all land on the while-you-were-gone
// card. The double offer rides the shared double 5/day category (one
// offer per span, the frozen 1-5/200 window band): the armed guard
// first, the HasCategoryBudget precheck second (one policy instance,
// one day bucket - precheck and consume can never disagree), the
// rewarded view third, and only the completed view books the earned
// gold a second time through the event ledger line - the wallet law
// (gold == start - slot costs + booked ledger lines) holds by
// construction. A capped tap refuses BEFORE the ad ever plays; an
// aborted view changes nothing and the retry stays open. Pure C#
// zero-RNG: the same absence span always settles byte-identical.
using System;
using System.Collections.Generic;
using System.Globalization;
using System.Text;
using System.Threading.Tasks;
using Sy.Platform;
namespace Sy.CrazyHospital
{
    public sealed class HospitalOffline
    {
        /// &summary&The offline double point id (T010): rides the shared

```

```

    /// double category through AdCategories.CategoryOfPoint authority
    /// (the g09_ prefix keeps id namespaces isolated, the
    /// g08/g10/g11/g13/g14 doubler precedents).</summary>
    public const string DoublePoint = "g09_offline_double";
    /// </summary>The night-desk walk slice: the nurse checks the hall
    /// once a second (a freed bed takes the next companion within the
    /// same second it freed in).</summary>
    public const double WalkSliceSeconds = 1.0;
    readonly HospitalStage _stage;
    readonly AdCapPolicy _adCap;
    readonly IAdsService _ads;
    readonly Func<string> _dayKey;
    readonly List<string> _funnel = new List<string>();
    readonly List<string> _summary = new List<string>();
    int _spanIndex;
    double _spanSeconds;
    double _simSeconds;
    int _tideArrivals;
    int _eventsRolledBefore;
    int _goldBeforeHc;
    int _earnHc;
    bool _armed;
    bool _claimed;
    public HospitalOffline(HospitalStage stage, AdCapPolicy adCap, IAdsService ads, Func<string> dayKey = null
    {
        _stage = stage;
        _adCap = adCap;
        _ads = ads;
        _dayKey = dayKey ?? AdCapPolicy.TodayKey;
        if (ads != null) ads.OnAdEvent += OnAdLifecycle;
    }
    // ---- live faces ----
    public HospitalStage Stage => _stage;
    public double SpanSeconds => _spanSeconds;
    public double SimSeconds => _simSeconds;
    public int TideArrivals => _tideArrivals;
    public int EarnHc => _earnHc;
    public bool OfferArmed => _armed;
    public bool Claimed => _claimed;
    public IReadOnlyList<string> Funnel => _funnel;
    public IReadOnlyList<string> SummaryLines => _summary;
    string SpanTag => "s" + _spanIndex;
    /// </summary>The night-tide fold: the extra walk-ins the x1.5
    /// offline-window traffic books over the natural street stream
    /// (the natural stream keeps rolling inside the walk itself).
    /// Deterministic off the settle-start traffic snapshot.</summary>
    public static int TideArrivalsFor(int trafficPerMinute, double simSeconds)
        => (int)Math.Ceiling(trafficPerMinute * simSeconds / 60.0
            * (HospitalData.NightEmergencyFactorBp - 10000) / 10000.0);
    /// </summary>Settles one absence span: caps at the frozen 12h,

```

```
/// folds to the 50% pace, front-loads the night tide queue, walks
/// the whole stage through the sim seconds with the night nurse on
/// the desk, then writes the while-you-were-gone summary and arms
/// the double offer once (only when the span earned something to
/// double). A zero absence books nothing at all: head line only,
/// no funnel, the offer never arms (the zero-span law).&lt;/summary&gt;
public void Settle(double absentSeconds)
{
    _spanIndex++;
    _claimed = false;
    _armed = false;
    _earnHc = 0;
    _tideArrivals = 0;
    _summary.Clear();
    _summary.Add(HospitalTexts.EnText(HospitalTexts.KeyOfflineSummary));
    if (absentSeconds &lt;= 0)
    {
        // the zero-span law: zero sim, zero funnel, head line only
        _spanSeconds = 0;
        _simSeconds = 0;
        return;
    }
    _spanSeconds = Math.Min(absentSeconds, HospitalData.OfflineCapHours * 3600.0);
    _simSeconds = _spanSeconds * HospitalData.OfflineRatePercent / 100.0;
    var intake = _stage.Intake;
    _tideArrivals = TideArrivalsFor(intake.TrafficPerMinute, _simSeconds);
    _eventsRolledBefore = _stage.Events.RollCount;
    _goldBeforeHc = intake.GoldHc;
    intake.AddStreetQueue(_tideArrivals); // the tide front-load: they lined up while you slept
    _funnel.Add("evt_offline_settle@" + SpanTag + ":"
        + _spanSeconds.ToString("0.#", CultureInfo.InvariantCulture) + "&gt;"
        + _simSeconds.ToString("0.#", CultureInfo.InvariantCulture)
        + ":tide" + _tideArrivals);
    double remaining = _simSeconds;
    while (remaining &gt; 0.0000001)
    {
        double step = Math.Min(WalkSliceSeconds, remaining);
        _stage.Advance(step);
        remaining -= step;
        // the night desk: a freed bed takes the next companion
        if (intake.OpenSlots &gt; 0 && intake.QueueCount &gt; 0)
            intake.TryRegisterQueueHead();
    }
    _earnHc = intake.GoldHc - _goldBeforeHc;
    _armed = _earnHc &gt; 0; // one offer per span, only when there is something to double
    BuildSummary();
}
void BuildSummary()
{
    var intake = _stage.Intake;
```

```

        _summary.Add(HospitalTexts.EnText(HospitalTexts.KeyNightTide) + " " + _tideArrivals
            + " EXTRA WALK-INS - THE HALL SLEPT FULL OF COMPANIONS");
        _summary.Add("away " + HoursStr(_spanSeconds) + "h -> " + HoursStr(_simSeconds) + "h at "
            + HospitalData.OfflineRatePercent + "% pace");
        int rolled = _stage.Events.RollCount - _eventsRolledBefore;
        if (rolled > 0 || _stage.Events.PendingCard != null)
            _summary.Add("the night desk kept busy: " + rolled + " event cards rolled, "
                + (_stage.Events.PendingCard != null ? "one is waiting on your desk" : "none left waiting"));
        _summary.Add("the street never emptied: " + intake.QueueCount + " still lining up outside");
        _summary.Add("gold " + HcStr(_goldBeforeHc) + " -> " + HcStr(intake.GoldHc)
            + " (" + (_earnHc >= 0 ? "+" : "") + HcStr(_earnHc) + ")");
    }
    /// <summary>The double offer claim (the shared double category,
    /// one offer per span): the armed guard first (a dead tap changes
    /// nothing - zero state, zero budget), the category precheck
    /// second, the rewarded view third; only the completed view books
    /// the earned gold a second time through the event ledger line.
    /// A capped tap refuses BEFORE the ad; an aborted view keeps the
    /// span armed for the retry.</summary>
    public async Task<bool> ClaimDouble()
    {
        if (!_armed) return false; // the dead tap: zero state, zero budget
        if (_adCap == null || _ads == null) return false;
        if (!_adCap.HasCategoryBudget(_dayKey(), AdCategories.CategoryOfPoint(DoublePoint)))
        {
            _funnel.Add("evt_ad_offer_capped@" + SpanTag);
            return false; // the capped tap refuses BEFORE the ad ever plays
        }
        _funnel.Add("evt_ad_offer@" + SpanTag);
        RewardResult r = await _ads.ShowRewarded(DoublePoint);
        if (r == null || !r.Completed) return false; // the aborted view: armed stays, the retry window stays open
        _stage.Intake.BookEventGoldHc(_earnHc); // the x2 injection rides the event ledger line - the wallet law h
        _funnel.Add("evt_offline_double@" + SpanTag + ":" + _earnHc);
        _armed = false;
        _claimed = true;
        return true;
    }
    /// <summary>evt_ad funnel (the registered-service lifecycle face):
    /// show/complete/fail of the double point land in order.</summary>
    void OnAdLifecycle(AdLifecycle e)
    {
        if (e.PointId != DoublePoint) return;
        _funnel.Add("evt_ad_" + e.Type.ToString().ToLowerInvariant() + "@" + SpanTag);
    }
    static string HoursStr(double seconds)
        => (seconds / 3600.0).ToString("0.##", CultureInfo.InvariantCulture);
    static string HcStr(long hc)
        => (hc / 100.0).ToString("0.##", CultureInfo.InvariantCulture);
    /// <summary>Canonical fingerprint - the zero-RNG parity instrument
    /// (the settle state plus the whole stage opening signature).</summary>

```

```

    public string Fingerprint()
    {
        var intake = _stage.Intake;
        var sb = new StringBuilder();
        sb.Append("span").Append(_spanIndex).Append('@')
            .Append(_spanSeconds.ToString("0.#", CultureInfo.InvariantCulture)).Append('&gt;');
            .Append(_simSeconds.ToString("0.#", CultureInfo.InvariantCulture));
        sb.Append("|tide").Append(_tideArrivals);
        sb.Append("|earn").Append(_earnHc);
        sb.Append("|armed").Append(_armed ? 1 : 0).Append(_claimed ? "C" : "-");
        sb.Append("|queue").Append(intake.QueueCount).Append(',').Append(intake.ArrivalTotal)
            .Append(',').Append(intake.PatientTotal).Append(',').Append(intake.CuredTotal);
        sb.Append("|fun").Append(_funnel.Count).Append("|sum").Append(_summary.Count);
        foreach (var s in _summary) sb.Append('#').Append(s);
        sb.Append("||").Append(_stage.OpeningSig());
        return sb.ToString();
    }
}

// HospitalPlaceholder.cs
// HospitalPlaceholder.cs - procedural placeholder faces for Crazy Hospital
// (G09-T004). Art-first iron rule (X027): when ArtRuntime holds the stem the
// real sprite wins, otherwise the face falls back to a procedural render -
// views render either way and no headless gate ever depends on art presence.
// The series accent is mint teal #6ED8C8 (the G09 brand face). The emblem
// family is the GREEN CROSS (the red-cross avoidance law: a fictional green
// cross marks every medical face, never a real emblem).
using System.Collections.Generic;
using Sy.Core;
using UnityEngine;
namespace Sy.CrazyHospital
{
    public static class HospitalPlaceholder
    {
        /// &lt;summary&gt;Series accent: mint teal #6ED8C8 (design-language face).&lt;/summary&gt;
        public static readonly Color AccentMint = new Color(110f / 255f, 216f / 255f, 200f / 255f);
        // ---- rank tints: candy clinic -&gt; cream -&gt; fresh mint -&gt; clinical
        // teal -&gt; sterile steel (the vertical facade warms as the hospital
        // ranks up) ----
        static readonly Color[] RankTints =
        {
            new Color(0.96f, 0.86f, 0.80f), // R1 candy clinic
            new Color(0.97f, 0.94f, 0.82f), // R2 warm cream
            new Color(0.78f, 0.94f, 0.88f), // R3 fresh mint
            new Color(0.62f, 0.82f, 0.86f), // R4 clinical teal
            new Color(0.52f, 0.62f, 0.72f), // R5 sterile steel
        };
        // ---- stems (art ships under these keys in S2.5; today they miss -&gt; fallback) ----
        public const string FacadeStem = "g09_facade";
        public const string PatientStem = "g09_patient";
    }
}

```

```

public const string DoctorStem = "g09_doctor";
public const string DeptStem = "g09_dept";
public const string EventCardStem = "g09_eventcard";
public const string PanelStem = "g09_panel";
static readonly Dictionary<string, Sprite> _cache = new Dictionary<string, Sprite>();
/// <summary>True when real art ships under the stem (S2.5 face - the
/// placeholder gates branch on this so art arrival never breaks them).</summary>
public static bool ArtPresent(string stem) => ArtRuntime.Get(stem) != null;
/// <summary>The shipped art sprite for a stem (null on the placeholder
/// path) - editor gates compare against this without touching Sy.Core.</summary>
public static Sprite ArtSprite(string stem) => ArtRuntime.Get(stem);
/// <summary>Hospital facade tinted by rank (1-5, out-of-range folds
/// to the R1 candy face). The vertical growth roll face. Key law: the
/// shipped catalog stems carry the _r rank suffix (art book sec7,
/// g09_facade_r1..r5) - a bare _&lt;rank> probe silently misses the
/// art and falls back to the tint block (the C1081 wiring bug this
/// resolver must never reintroduce; ArtCheck S3 owns the law).</summary>
public static Sprite Facade(int rank)
{
    int i = rank - 1;
    if (i < 0 || i >= RankTints.Length) i = 0;
    return Face(FacadeStem + "_r" + (i + 1), RankTints[i], 64);
}
public static Sprite Patient() => Face(PatientStem, new Color(0.92f, 0.80f, 0.88f), 32);
public static Sprite Doctor() => Face(DoctorStem, new Color(0.80f, 0.88f, 0.95f), 32);
/// <summary>The department face: art-first, else a green-cross block
/// (the fictional emblem, red-cross avoidance law).</summary>
public static Sprite Dept()
{
    Sprite art = ArtRuntime.Get(DeptStem);
    if (art != null) return art;
    Sprite cached;
    if (_cache.TryGetValue(DeptStem, out cached) && cached != null) return cached;
    Sprite made = ProceduralCross(32, AccentMint, new Color(0.94f, 0.98f, 0.97f));
    _cache[DeptStem] = made;
    return made;
}
public static Sprite EventCard() => Face(EventCardStem, new Color(0.95f, 0.90f, 0.66f), 48);
public static Sprite Panel() => Face(PanelStem, new Color(0.14f, 0.18f, 0.20f, 0.92f), 64);
/// <summary>Art-first with null-safe procedural fallback (X027 iron rule).</summary>
public static Sprite Face(string stem, Color color, int px)
{
    Sprite art = ArtRuntime.Get(stem);
    if (art != null) return art;
    string key = stem + "#" + px;
    Sprite cached;
    if (_cache.TryGetValue(key, out cached) && cached != null) return cached;
    Sprite made = ProceduralSprite(color, px);
    _cache[key] = made;
    return made;
}

```

```
}
/// <summary>Procedural color block: an uncompressed readable px*px square.</summary>
public static Sprite ProceduralSprite(Color color, int px)
{
    var tex = new Texture2D(px, px, TextureFormat.RGBA32, false);
    var pixels = new Color[px * px];
    for (int i = 0; i < pixels.Length; i++) pixels[i] = color;
    tex.SetPixels(pixels);
    tex.Apply(false, false);
    tex.name = "sy_procedural_" + px;
    var sprite = Sprite.Create(tex, new Rect(0, 0, px, px), new Vector2(0.5f, 0.5f), px);
    sprite.name = tex.name;
    return sprite;
}
/// <summary>The green-cross procedural emblem: a plus-shaped cross in
/// the accent color on a soft block (the fictional medical mark - a
/// pure pixel draw, stable forever for gate pixel reads).</summary>
public static Sprite ProceduralCross(int px, Color cross, Color background)
{
    var tex = new Texture2D(px, px, TextureFormat.RGBA32, false);
    var pixels = new Color[px * px];
    int arm = px / 8;          // arm thickness (>=1 on any px >= 8)
    if (arm < 1) arm = 1;
    int mid = px / 2;
    for (int y = 0; y < px; y++)
    {
        for (int x = 0; x < px; x++)
        {
            bool inCross = (x >= mid - arm && x <= mid + arm - 1) || (y >= mid - arm &&
            pixels[y * px + x] = inCross ? cross : background;
        }
    }
    tex.SetPixels(pixels);
    tex.Apply(false, false);
    tex.name = "sy_green_cross_" + px;
    var sprite = Sprite.Create(tex, new Rect(0, 0, px, px), new Vector2(0.5f, 0.5f), px);
    sprite.name = tex.name;
    return sprite;
}
/// <summary>The rank tint for a hospital rank (1-5) - the visual
/// progress face for checks.</summary>
public static Color TintForRank(int rank)
{
    int i = rank - 1;
    if (i < 0 || i >= RankTints.Length) i = 0;
    return RankTints[i];
}
}
// HospitalRank.cs
```

```

// HospitalRank.cs - the rank-review gate engine (G09-T008, SYS05/SYS06).
// The five-rank ladder R1-R5 walks four dual-track gates (numeric pack
// section three): the hard metrics (departments / cure rate / reputation)
// AND the inspection score must hold together - a promotion is a new
// gameplay ticket, never a wall. A failed review only arms the
// rectification comedy (the review expert rides the catapult into the
// hospital's own ward) and re-arms the next review window at the frozen
// inspection cadence (180s, the events-engine cycle - zero new numbers).
// The settlement face is the frozen closed set: pass swings reputation
// +3, fail -3 (HospitalData.ReviewRepSwing); the E01 review bonus folds
// +3 into the score, E04 folds +2 (the events engine already books
// both). The reputation band is the second numeric line (section four):
// band changes book their visible face flags (banner growth at the top
// band, bad-review comedy at the bottom) plus the evt_band_change
// funnel line - the street flow itself already rides the frozen
// reputation function (TrafficAt), so the band face never re-models it.
// The engine keeps its OWN clock and funnel; the opening signature of
// the stage stays byte-identical until a review resolves (identity
// law). A resolved promotion folds the swap into both hosts inside the
// settle: the intake economy row (PromoteToRank) and the events stage
// pool (SetRank) - one verdict, three faces agree (the T008 host
// wiring). Pure C#, no MonoBehaviour.
using System;
using System.Collections.Generic;
namespace Sy.CrazyHospital
{
    /// <summary>One review settlement (the dual-track verdict).</summary>
    public struct ReviewVerdict
    {
        public bool pass;
        public bool hardOk;
        public bool scoreOk;
    }
    /// <summary>SYS05/SYS06: the rank-review gate + reputation band
    /// engine. The host wires Advance(dt); headless gates drive the
    /// static solver and the gate faces directly.</summary>
    public sealed class HospitalRank
    {
        // ---- the frozen review laws (closed set, zero new numbers) ----
        public const double ReviewCooldownSeconds = HospitalEvents.CadenceSeconds; // re-review = the next inspection cy
        public const int ScoreBonus = HospitalData.ReviewRepSwing; // the E01 pass folds +3 into the score
        readonly HospitalIntake _intake;
        readonly HospitalEvents _events;
        readonly List<string> _funnel = new List<string>();
        double _t;
        double _cooldownUntil = -1; // sim time until the next review window opens (-1 = open)
        int _rank = 1;
        int _promotions;
        int _failedReviews;
        int _lastVerdictRank; // rank the last review ran under (0 = never)
    }
}

```

```

bool _lastPass;
bool _ceremonyArmed;          // pass -&gt; the stamp ceremony (fireworks + banner rain)
bool _remediationArmed;       // fail -&gt; the rectification comedy face
bool _badReviewComedyArmed;    // band entry into rep_bad
bool _bannerGrowthArmed;      // band entry into rep_banner
string _bandId = "";
public HospitalRank(HospitalIntake intake, HospitalEvents events, int startRank = 1)
{
    _intake = intake;
    _events = events;
    _rank = startRank &lt; 1 ? 1 : startRank;
    _bandId = BandFace().id;
}
// ---- live faces ----
public double SimTime =&gt; _t;
public int Rank =&gt; _rank;
public int Promotions =&gt; _promotions;
public int FailedReviews =&gt; _failedReviews;
public int LastVerdictRank =&gt; _lastVerdictRank;
public bool LastPass =&gt; _lastPass;
public bool CeremonyArmed =&gt; _ceremonyArmed;
public bool RemediationArmed =&gt; _remediationArmed;
public bool BadReviewComedyArmed =&gt; _badReviewComedyArmed;
public bool BannerGrowthArmed =&gt; _bannerGrowthArmed;
public string BandId =&gt; _bandId;
public IReadOnlyList<string> Funnel =&gt; _funnel;
public bool ReviewWindowOpen =&gt; _t &gt;= _cooldownUntil;
/// &lt;summary&gt;The gate the current rank faces (null at R5: the
/// ladder is complete, the top-grade stamp is the ending face).&lt;/summary&gt;
public ReviewGateDef CurrentGate
{
    get
    {
        foreach (var g in HospitalData.Gates())
            if (g.fromRank == _rank) return g;
        return null;
    }
}
/// &lt;summary&gt;The live inspection score: the events-engine delta
/// plus the armed E01 review bonus (the closed set: E04 +-2,
/// E01 pass +3).&lt;/summary&gt;
public int InspectScore
    =&gt; (_events != null ? _events.InspectScoreDelta : 0)
    + (_events != null & & _events.ReviewBonusPending ? ScoreBonus : 0);
/// &lt;summary&gt;Live hard-metric faces (the host already owns the
/// sim; the solver reads, never models).&lt;/summary&gt;
public int DeptsOf() =&gt; _intake != null ? _intake.OpenSlots : 0;
public int ReputationOf() =&gt; _intake != null ? _intake.Reputation : 0;
public int CureRatePctOf()
{

```

```

        if (_intake == null || _intake.PatientTotal &lt;= 0) return 0;
        return _intake.CuredTotal * 100 / _intake.PatientTotal;
    }
    /// &lt;summary&gt;The reputation band (the frozen section-four
    /// partition; the street flow rides TrafficAt, never re-model).&lt;/summary&gt;
    public ReputationBand BandFace()
        =&gt; _intake != null ? HospitalData.BandFor(_intake.Reputation)
        : HospitalData.BandFor(HospitalData.ReputationMin);
    /// &lt;summary&gt;Static dual-track solver (the headless-gate face:
    /// verdict math is pure, no runtime state).&lt;/summary&gt;
    public static ReviewVerdict Solve(ReviewGateDef gate, int depts, int cureRatePct, int reputation, int inspectSco
    {
        ReviewVerdict v;
        v.hardOk = gate != null
            &amp;&amp; depts &gt;= gate.minDepts
            &amp;&amp; cureRatePct &gt;= gate.minCureRatePct
            &amp;&amp; reputation &gt;= gate.minReputation;
        v.scoreOk = inspectScore &gt;= 0;    // a net non-negative inspection record
        v.pass = gate != null &amp;&amp; v.hardOk &amp;&amp; v.scoreOk;    // dual-track law: both tracks together
        return v;
    }
    /// &lt;summary&gt;Requests a review against the current gate: refuses
    /// while the cooldown window holds (fail = the next inspection
    /// cycle) or at R5. On resolve books the settlement (rep swing
    /// +-3), the ceremony / remediation flags and the evt_promote /
    /// evt_inspect funnel lines. A pass promotes the ladder AND folds
    /// the swap into both hosts inside the settle: the intake economy
    /// row and the events stage pool read the new rank from the next
    /// tick on (one verdict, three faces agree).&lt;/summary&gt;
    public ReviewVerdict RequestReview()
    {
        var gate = CurrentGate;
        if (gate == null || !ReviewWindowOpen) { var none = new ReviewVerdict(); return none; }
        ReviewVerdict v = Solve(gate, DeptsOf(), CureRatePctOf(), ReputationOf(), InspectScore);
        _lastVerdictRank = _rank;
        _lastPass = v.pass;
        if (_intake != null)
            _intake.SetReputation(_intake.Reputation + (v.pass ? HospitalData.ReviewRepSwing : -HospitalData.ReviewR
        if (v.pass)
        {
            _rank = gate.toRank;
            if (_intake != null) _intake.PromoteToRank(gate.toRank);    // the host fold: the economy row swaps with
            if (_events != null) _events.SetRank(gate.toRank);        // the host fold: the stage pool widens with
            _promotions++;
            _ceremonyArmed = true;
            _funnel.Add("evt_promote@r" + gate.toRank + ":" + gate.reviewEventId);
        }
        else
        {
            _failedReviews++;

```

```
        _remediationArmed = true;
        _cooldownUntil = _t + ReviewCooldownSeconds;    // re-review = the next inspection cycle
        _funnel.Add("evt_inspect@r" + gate.fromRank + ":" + gate.reviewEventId);
    }
    return v;
}

/// <summary>Advances the engine clock and tracks the reputation
/// band face: a band change books the visible comedy flags and
/// the evt_band_change funnel line (banner growth at the top
/// band, bad-review comedy at the bottom - the two-way face of
/// the second numeric line).</summary>
public void Advance(double dt)
{
    if (dt <= 0) return;
    _t += dt;
    var band = BandFace();
    if (band.id != _bandId)
    {
        _bandId = band.id;
        _badReviewComedyArmed = band.id == "rep_bad";
        _bannerGrowthArmed = band.id == "rep_banner";
        _funnel.Add("evt_band_change@" + band.id);
    }
}

/// <summary>Gate faces (headless checks drive the engine without
/// a live intake/events pair).</summary>
public void ArmForGate(int rank, int reputation)
{
    _rank = rank < 1 ? 1 : rank;
    if (_intake != null) _intake.SetReputation(reputation);
    _cooldownUntil = -1;
}

/// <summary>Canonical fingerprint: the zero-RNG parity
/// instrument (same script = same bytes).</summary>
public string Fingerprint()
{
    var sb = new System.Text.StringBuilder();
    sb.Append("t=").Append(_t.ToString("0.###"));
    sb.Append("|rank=").Append(_rank);
    sb.Append("|promo=").Append(_promotions);
    sb.Append("|fail=").Append(_failedReviews);
    sb.Append("|last=").Append(_lastVerdictRank).Append('/').Append(_lastPass ? 1 : 0);
    sb.Append("|score=").Append(InspectScore);
    sb.Append("|cd=").Append(_cooldownUntil.ToString("0.###"));
    sb.Append("|flags=").Append(_ceremonyArmed ? 1 : 0).Append(_remediationArmed ? 1 : 0)
        .Append(_badReviewComedyArmed ? 1 : 0).Append(_bannerGrowthArmed ? 1 : 0);
    sb.Append("|band=").Append(_bandId);
    sb.Append("|fun=").Append(_funnel.Count);
    return sb.ToString();
}
```

```

    }
}
// HospitalSave.cs
// HospitalSave.cs - the save system (G09-T014, G06SaveSystem / G12SaveSystem
// paradigm): single-key JSON persistence through the Sy.Core save service.
// Local storage is the only authority (red line 12, zero-server doctrine).
//
// Faces:
// - HospitalProfile: the save payload model. Only earned state is stored -
//   gold, rank, reputation, the department slot choice history (replayable),
//   the doctor card collection in hire order, the three-volume album rows,
//   research spend and the booked event chain flags. Duty roster state is
//   NEVER stored: it derives from the collection + open slot count on every
//   SyncRoster call, so a drifted duty row is structurally impossible
//   (derive-live zero-drift doctrine, G12 unlock-face sibling).
// - HospitalSaveSystem: four-state load (NewGame / Loaded / Migrated /
//   CorruptedReset). Version-migration skeleton: a payload without the
//   version key migrates by filling defaults for missing fields; a payload
//   from a newer build degrades to a fresh profile instead of guessing at
//   its schema. Corrupted rows drop with warnings, never fatal - a save can
//   brick a run only by refusing to load, never by throwing.
// - HospitalSaveSchema: editor-check verdict faces (pure C#, zero platform
//   coupling beyond ISaveService itself) so the Editor check rides these
//   faces instead of naming Sy.Core types directly (G12SaveSchema
//   paradigm). Faces mirror the save DoD: round-trip + four states +
//   migration defaults + tamper clamps + NG+ carry reconciliation against
//   HospitalEnding.SolveCarry / ApplyCarry.
using System;
using System.Collections.Generic;
using Sy.Core;
namespace Sy.CrazyHospital
{
    /// <summary>Outcome of a save-slot load (G06 four-state paradigm).</summary>
    public enum HospitalLoadStatus
    {
        NewGame,          // slot empty: fresh dean, nothing written yet
        Loaded,           // current-version payload applied as-is
        Migrated,         // older payload upgraded in place (defaults filled)
        CorruptedReset    // unreadable or future-version payload: profile reset
    }
    /// <summary>
    /// The G09 save payload (task T014): every earned face of the S2
    /// runtime in one serializable bag. Slot choices are the replayable
    /// history (pick WHAT opened, in open order) - the slot count, the
    /// last choice and the duty roster all derive from it live.
    /// </summary>
    public sealed class HospitalProfile
    {
        public int GoldHc;                // the wallet (gold hundredths)
        public int Rank = 1;              // R1..R5 ladder state
    }
}

```

```

    public int Reputation;                // 0-100 street fame
    public List<int> SlotChoices = new List<int>(); // department choice history
    public int PullCount;                // doctor card draws taken
    public int DupeTotal;                // duplicate cards folded
    public int ResearchSpendHc;          // gold sunk into wonder drugs
    public List<string> DoctorIds = new List<string>(); // hire-order collection
    public List<string> AlbumIds = new List<string>(); // the three-volume unlocked rows
    public List<string> EventFlags = new List<string>(); // booked event chain flags
    /// <summary>Derived live from the replay history, never stored twice.</summary>
    public int OpenSlots => SlotChoices.Count;
}
/// <summary>
/// Single-key JSON persistence through the Sy.Core save service
/// (local authority, red line 12). All reads clamp to the frozen
/// schema laws (HospitalData) - a tampered payload can only ever
/// produce a schema-legal profile.
/// </summary>
public static class HospitalSaveSystem
{
    public const string Slot = "g09_main";
    public const string VersionKey = "g09v";
    public const int CurrentVersion = 1;
    /// <summary>Serialize the profile into the single-key JSON payload.</summary>
    public static string Serialize(HospitalProfile profile)
    {
        if (profile == null) throw new ArgumentNullException(nameof(profile));
        var choices = new List<object>();
        if (profile.SlotChoices != null)
            foreach (var c in profile.SlotChoices) choices.Add((long)c);
        var docs = new List<object>();
        if (profile.DoctorIds != null)
            foreach (var id in profile.DoctorIds) docs.Add(id);
        var albums = new List<object>();
        if (profile.AlbumIds != null)
            foreach (var id in profile.AlbumIds) albums.Add(id);
        var flags = new List<object>();
        if (profile.EventFlags != null)
            foreach (var id in profile.EventFlags) flags.Add(id);
        var data = new Dictionary<string, object>
        {
            { VersionKey, (long)CurrentVersion },
            { "gold", (long)profile.GoldHc },
            { "rank", (long)profile.Rank },
            { "rep", (long)profile.Reputation },
            { "choices", choices },
            { "pulls", (long)profile.PullCount },
            { "dupes", (long)profile.DupeTotal },
            { "rshc", (long)profile.ResearchSpendHc },
            { "docs", docs },
            { "albums", albums },
        }
    }
}

```

```
        { "flags", flags },
    };
    return MiniJson.Serialize(data);
}
/// <summary>Write the profile through the save service (local authority).</summary>
public static void Persist(ISaveService save, HospitalProfile profile, string slot = Slot)
{
    if (save == null) return;
    save.Save(slot, Serialize(profile));
}
/// <summary>
/// Load the slot into the given profile. Never throws on bad data:
/// corrupted or future-version payloads reset the profile to fresh
/// defaults. NewGame and CorruptedReset both reset deterministically
/// first, so a stale caller-side profile never survives either fresh
/// outcome.
/// </summary>
public static HospitalLoadStatus LoadInto(ISaveService save, HospitalProfile profile, string slot = Slot)
{
    if (save == null) throw new ArgumentNullException(nameof(save));
    if (profile == null) throw new ArgumentNullException(nameof(profile));
    string raw;
    try
    {
        raw = save.Load(slot);
    }
    catch (Exception e)
    {
        // the service itself chokes on a corrupted payload - degrade, never crash the boot
        Logger.Warn("G09.Save", "corrupt payload at service level, resetting slot: " + e.Message);
        return Fresh(profile);
    }
    if (string.IsNullOrEmpty(raw))
    {
        ResetProfile(profile);
        return HospitalLoadStatus.NewGame;
    }
    Dictionary<string, object> data;
    try
    {
        data = MiniJson.Deserialize(raw) as Dictionary<string, object>;
    }
    catch (Exception e)
    {
        Logger.Warn("G09.Save", "corrupt payload, resetting slot: " + e.Message);
        return Fresh(profile);
    }
    if (data == null)
    {
        Logger.Warn("G09.Save", "non-dict payload, resetting slot");
    }
}
```

```
        return Fresh(profile);
    }
    int version = ReadInt(data, VersionKey, 0);
    if (version > CurrentVersion)
    {
        Logger.Warn("G09.Save", "future version " + version + " > " + CurrentVersion + ", resetting slot");
        return Fresh(profile);
    }
    profile.GoldHc = ClampMin(ReadInt(data, "gold", 0), 0); // wallet law: spend refuses before negat
    profile.Rank = Clamp(ReadInt(data, "rank", 1), 1, HospitalData.RankCount);
    profile.Reputation = Clamp(ReadInt(data, "rep", 0),
        HospitalData.ReputationMin, HospitalData.ReputationMax);
    profile.SlotChoices = ReadInts(data, "choices");
    profile.PullCount = ClampMin(ReadInt(data, "pulls", 0), 0);
    profile.DupeTotal = ClampMin(ReadInt(data, "dupes", 0), 0);
    profile.ResearchSpendHc = ClampMin(ReadInt(data, "rshc", 0), 0);
    profile.DoctorIds = ReadIds(data, "docs");
    profile.AlbumIds = ReadIds(data, "albums");
    profile.EventFlags = ReadIds(data, "flags");
    return version < CurrentVersion ? HospitalLoadStatus.Migrated : HospitalLoadStatus.Loaded;
}
static HospitalLoadStatus Fresh(HospitalProfile profile)
{
    ResetProfile(profile);
    return HospitalLoadStatus.CorruptedReset;
}
static void ResetProfile(HospitalProfile profile)
{
    profile.GoldHc = 0;
    profile.Rank = 1;
    profile.Reputation = 0;
    profile.SlotChoices = new List<int>();
    profile.PullCount = 0;
    profile.DupeTotal = 0;
    profile.ResearchSpendHc = 0;
    profile.DoctorIds = new List<string>();
    profile.AlbumIds = new List<string>();
    profile.EventFlags = new List<string>();
}
static int Clamp(int v, int lo, int hi) => v < lo ? lo : (v > hi ? hi : v);
static int ClampMin(int v, int lo) => v < lo ? lo : v;
static int ReadInt(Dictionary<string, object> data, string key, int fallback)
{
    if (!data.TryGetValue(key, out var value) || value == null) return fallback;
    try { return Convert.ToInt32(Convert.ToDouble(value)); }
    catch { return fallback; }
}
static List<int> ReadInts(Dictionary<string, object> data, string key)
{
    var list = new List<int>();
```

```
        if (!data.TryGetValue(key, out var value) || !(value is System.Collections.IEnumerable rows)) return list;
        foreach (var row in rows)
        {
            int v;
            try { v = Convert.ToInt32(Convert.ToDouble(row)); }
            catch
            {
                Logger.Warn("G09.Save", "non-int choice row, dropped");
                continue;
            }
            if (v < 0)
            {
                Logger.Warn("G09.Save", "negative choice row, dropped");
                continue;
            }
            list.Add(v);
        }
        return list;
    }

    /// <summary>Id rows: null / empty / duplicate entries drop with a
    /// warning (a save can never inflate a collection face).</summary>
    static List<string> ReadIds(Dictionary<string, object> data, string key)
    {
        var list = new List<string>();
        var seen = new HashSet<string>();
        if (!data.TryGetValue(key, out var value) || !(value is System.Collections.IEnumerable rows)) return list;
        foreach (var row in rows)
        {
            string id = row as string;
            if (string.IsNullOrEmpty(id))
            {
                Logger.Warn("G09.Save", key + " row without id, dropped");
                continue;
            }
            if (!seen.Add(id))
            {
                Logger.Warn("G09.Save", key + " duplicate row, dropped: " + id);
                continue;
            }
            list.Add(id);
        }
        return list;
    }
}

/// <summary>
/// Editor-check verdict faces for the save domain (G12SaveSchema
/// paradigm): pure C#, EditMode-pinnable, mirroring the save DoD in
/// machine-readable form so the SaveCheck gate owns round-trip + four
/// states + migration defaults + tamper clamps + the NG+ carry
/// reconciliation against HospitalEnding.
```

```
/// <summary>
public static class HospitalSaveSchema
{
    /// <summary>In-memory ISaveService stub (check path only, zero platform coupling).</summary>
    sealed class MemorySaveService : ISaveService
    {
        readonly Dictionary<string, string> slots = new Dictionary<string, string>();
        public void Save(string slot, string payloadJson) { slots[slot] = payloadJson; }
        public string Load(string slot)
        {
            string v;
            return slots.TryGetValue(slot, out v) ? v : null;
        }
        public SyncResult CloudSync() { return new SyncResult(false, null); }
    }
    static HospitalProfile MakeSeededProfile()
    {
        return new HospitalProfile
        {
            GoldHc = 12345,
            Rank = 3,
            Reputation = 62,
            SlotChoices = new List<int> { 0, 2, 1 },
            PullCount = 7,
            DupeTotal = 2,
            ResearchSpendHc = 350,
            DoctorIds = new List<string> { "doc_threesec", "doc_wangdali" },
            AlbumIds = new List<string> { "dis_hiccup_bug", "drug_hiccup_cola" },
            EventFlags = new List<string> { HospitalEvents.ChainReviewBonus, HospitalEvents.ChainBanner },
        };
    }
    /// <summary>Serialize - persist - reload keeps every field and re-serializes identically.</summary>
    public static bool RoundTripPreservesPayload()
    {
        var save = new MemorySaveService();
        var seeded = MakeSeededProfile();
        string once = HospitalSaveSystem.Serialize(seeded);
        HospitalSaveSystem.Persist(save, seeded);
        var back = new HospitalProfile();
        if (HospitalSaveSystem.LoadInto(save, back) != HospitalLoadStatus.Loaded) return false;
        if (HospitalSaveSystem.Serialize(back) != once) return false;
        return back.GoldHc == 12345 && back.Rank == 3 && back.Reputation == 62
            && back.OpenSlots == 3 && back.PullCount == 7 && back.DupeTotal == 2
            && back.ResearchSpendHc == 350
            && back.DoctorIds.Count == 2 && back.DoctorIds[0] == "doc_threesec"
            && back.AlbumIds.Count == 2 && back.AlbumIds[0] == "dis_hiccup_bug"
            && back.EventFlags.Count == 2;
    }
    /// <summary>Empty slot = NewGame, and the reset wipes any stale caller-side rows.</summary>
    public static bool StateNewGameOnEmptySlot()
```

```

{
    var save = new MemorySaveService();
    var stale = MakeSeededProfile();
    if (HospitalSaveSystem.LoadInto(save, stale) != HospitalLoadStatus.NewGame) return false;
    return stale.GoldHc == 0 && stale.Rank == 1 && stale.OpenSlots == 0
        && stale.DoctorIds.Count == 0 && stale.AlbumIds.Count == 0 && stale.EventFlags.C
}
/// <summary>Version-less payload (pre-version build) migrates by filling defaults.</summary>
public static bool StateMigratedOnVersionlessPayload()
{
    var save = new MemorySaveService();
    save.Save(HospitalSaveSystem.Slot,
        "{ \"gold\":900, \"rank\":2, \"rep\":40, \"choices\":[0], \"docs\":[ \"doc_wangdali\" ], \"albums\":[ ], \"flags\"
    var profile = new HospitalProfile();
    if (HospitalSaveSystem.LoadInto(save, profile) != HospitalLoadStatus.Migrated) return false;
    // the missing keys (pulls / dupes / rshc) migrate to their defaults
    return profile.GoldHc == 900 && profile.Rank == 2 && profile.Reputation == 40
        && profile.OpenSlots == 1 && profile.PullCount == 0 && profile.DupeTotal == 0
        && profile.ResearchSpendHc == 0
        && profile.DoctorIds.Count == 1 && profile.DoctorIds[0] == "doc_wangdali";
}
/// <summary>Unparseable payload degrades to a fresh profile instead of crashing the boot.</summary>
public static bool StateCorruptedResetOnGarbage()
{
    var save = new MemorySaveService();
    save.Save(HospitalSaveSystem.Slot, "{ not json at all");
    var stale = MakeSeededProfile();
    if (HospitalSaveSystem.LoadInto(save, stale) != HospitalLoadStatus.CorruptedReset) return false;
    return stale.GoldHc == 0 && stale.OpenSlots == 0 && stale.DoctorIds.Count == 0;
}
/// <summary>A payload from a newer build resets rather than guessing at its schema.</summary>
public static bool StateCorruptedResetOnFutureVersion()
{
    var save = new MemorySaveService();
    save.Save(HospitalSaveSystem.Slot, "{ \"g09v\":99, \"gold\":1}");
    var profile = new HospitalProfile();
    return HospitalSaveSystem.LoadInto(save, profile) == HospitalLoadStatus.CorruptedReset
        && profile.GoldHc == 0;
}
/// <summary>A tampered payload can only ever produce a schema-legal
/// profile: the frozen clamps hold and collection rows dedupe.</summary>
public static bool TamperClampsToSchemaLaws()
{
    var save = new MemorySaveService();
    save.Save(HospitalSaveSystem.Slot,
        "{ \"g09v\":1, \"gold\":-500, \"rank\":9, \"rep\":250, \"choices\":[ -1,2 ], \"pulls\":[ -4, \"
        + \"docs\":[ \"doc_wangdali\", \"doc_wangdali\", \"\" ], \"albums\":[ \"dis_monday\" ], \"flags\":[ ] }");
    var profile = new HospitalProfile();
    if (HospitalSaveSystem.LoadInto(save, profile) != HospitalLoadStatus.Loaded) return false;
    return profile.GoldHc == 0

```

```
&& profile.Rank == HospitalData.RankCount
&& profile.Reputation == HospitalData.ReputationMax
&& profile.OpenSlots == 1 && profile.SlotChoices[0] == 2
&& profile.PullCount == 0
&& profile.DoctorIds.Count == 1 && profile.DoctorIds[0] == "doc_wangdali";
}
/// <summary>NG+ carry reconciliation (task DoD: the inherit face
/// matches HospitalEnding): the carry pack folded from the saved
/// profile's faces keeps gold 20% floored and the album rows, rank
/// and event flags reset by simply NOT being carried, and applying
/// the pack onto a fresh album replants the same ids with zero drift.</summary>
public static bool NgPlusCarryMatchesEndingFaces()
{
    var save = new MemorySaveService();
    var seeded = MakeSeededProfile();
    HospitalSaveSystem.Persist(save, seeded);
    var back = new HospitalProfile();
    if (HospitalSaveSystem.LoadInto(save, back) != HospitalLoadStatus.Loaded) return false;
    // fresh run faces: album replanted from the saved rows, doctor desk null
    // (the doctor id leg rides the profile round-trip, the desk's own
    // InheritFrom idempotence is DoctorCheck territory)
    var freshAlbum = new HospitalAlbum(null);
    var carry = HospitalEnding.SolveCarry(back.GoldHc, AlbumFromProfile(back), null);
    if (carry.GoldCarryHc != HospitalEnding.SolveGoldCarryHc(back.GoldHc)) return false;
    if (carry.GoldCarryHc != back.GoldHc * HospitalEnding.GoldCarryPercent / 100) return false;
    if (carry.DoctorIds == null || carry.DoctorIds.Count != 0) return false;
    var ending = new HospitalEnding();
    ending.ApplyCarry(carry, freshAlbum, null);
    // the replanted album rows match the saved rows exactly (zero drift)
    if (freshAlbum.Unlocked.Count != back.AlbumIds.Count) return false;
    for (int i = 0; i < back.AlbumIds.Count; i++)
        if (freshAlbum.Unlocked[i] != back.AlbumIds[i]) return false;
    // the ending books its one carry line, and the once-guard facade face stays untouched
    if (ending.FacadeLit) return false;
    // the fresh run's own save profile is a new-game identity: rank / flags never carry
    var freshProfile = new HospitalProfile();
    if (freshProfile.Rank != 1 || freshProfile.EventFlags.Count != 0) return false;
    return true;
}
/// <summary>Duty roster state derives live from the collection +
/// slot count (SyncRoster), never from the save payload - a drifted
/// duty row is structurally impossible.</summary>
public static bool DutyRosterDerivesZeroDrift()
{
    var desk = new HospitalDoctors(null, null, () => "gate_day");
    desk.InheritFrom(new List<string> { "doc_threesecond", "doc_wangdali" });
    desk.SyncRoster(1);
    if (desk.Roster.Count != 1) return false;
    desk.SyncRoster(4);
    if (desk.Roster.Count != 2) return false;
```

```

        // the save payload carries no roster field at all: the serialized
        // seed contains no duty face, only the collection ids
        string json = HospitalSaveSystem.Serialize(MakeSeededProfile());
        return json.Contains("docs") && !json.Contains("duty") && !json.Contains("roster");
    }
    static HospitalAlbum AlbumFromProfile(HospitalProfile profile)
    {
        var album = new HospitalAlbum(null);
        if (profile != null && profile.AlbumIds != null) album.InheritFrom(profile.AlbumIds);
        return album;
    }
}

// HospitalShareCard.cs
// HospitalShareCard.cs - the top-grade stamp certificate share card
// (G09-T011 prefix, SYS08 / R5 ending face): the brag card a stamped
// dean shows off - the grade title, four stat lines, the honor-wall
// photo row and the challenge footer - composed through the shared
// AutoCardComposer face (Sy.Social; template key g09_sharecard:
// missing art falls back to the procedural dark card, headless stays
// green). Every delivered share books evt_album_share; a compose that
// never lands books nothing (the funnel only ever counts real shares).
// The honor-wall law is closed-set: one stamp cell (tier = rank 1-5)
// plus one cell per album volume (tier = rate/20 floored: 0 empty,
// 100% gold). Pure C# + the shared composer, zero RNG, zero scene
// state, every copy routed through HospitalTexts keys.
using System.Collections.Generic;
using Sy.Social;
using UnityEngine;
namespace Sy.CrazyHospital
{
    public sealed class HospitalShareCard
    {
        public const string TemplateKey = "g09_sharecard";
        public const string EventId = "evt_album_share";
        public const string QueryPrefix = "g09";
        public const int WallCells = 4; // stamp + three album volumes
        public const int PinnedComposeBudgetBytes = 8 * 1024 * 1024; // canary budget pins the 1080x1440 tier
        readonly List<string> _funnel = new List<string>();
        public IReadOnlyList<string> Funnel => _funnel;
        /// <summary>Honor-wall tier solver (closed set): rate 0-19 -> 0
        /// (empty cell), each +20% one tier up, 100% = gold tier 5,
        /// out-of-range rates clamp.</summary>
        public static int SolveWallTier(int ratePct)
            => Mathf.Clamp(ratePct / 20, 0, 5);
        /// <summary>The collection-depth face of the share payload: the
        /// sum of wall tiers (the query carries it as vols).</summary>
        public static int VolumeTierSum(int[] volumeRates)
        {
            int n = 0;

```

```

        if (volumeRates != null) for (int i = 0; i < volumeRates.Length && i < 3; i++) n += SolveWallT
        return n;
    }
    /// <summary>Pure spec solver (the headless face): title = the
    /// grade name of the rank, lines = rep / cure / depts / on-duty,
    /// footer = the challenge, wall = stamp + three volume cells in
    /// pack order (diseases / doctors / drugs).</summary>
    public static ShareCard SolveSpec(int rank, int reputation, int cureRatePct,
        int depts, int doctorsOnDuty, int[] volumeRates)
    {
        rank = Mathf.Clamp(rank, 1, 5);
        var rates = volumeRates ?? new int[0];
        var cells = new CardCell[WallCells];
        cells[0] = new CardCell { Tier = rank, Stack = 1 };
        for (int i = 0; i < 3; i++)
            cells[i + 1] = new CardCell { Tier = rates.Length > i ? SolveWallTier(rates[i]) : 0, Stack = 1 };
        var lines = new[]
        {
            HospitalTexts.EnText(HospitalTexts.KeyStatReputation) + ": " + reputation,
            HospitalTexts.EnText(HospitalTexts.KeyStatCureRate) + ": " + Mathf.Max(0, cureRatePct) + "%",
            HospitalTexts.EnText(HospitalTexts.KeyStatDepts) + ": " + Mathf.Max(0, depts),
            HospitalTexts.EnText(HospitalTexts.KeyDutyRoster) + ": " + Mathf.Max(0, doctorsOnDuty),
        };
        var query = QueryPrefix + "&rank=" + rank + "&rep=" + reputation
            + "&cure=" + Mathf.Max(0, cureRatePct) + "&depts=" + Mathf.Max(0, depts)
            + "&docs=" + Mathf.Max(0, doctorsOnDuty) + "&vols=" + VolumeTierSum(rates);
        var card = AutoCardComposer.Card(TemplateKey,
            HospitalTexts.EnText("rank_" + rank), lines,
            HospitalTexts.EnText(HospitalTexts.KeyShareChallenge), query);
        card.Formation = cells;
        card.FormationCols = WallCells;
        card.FormationRows = 1;
        return card;
    }
    /// <summary>Live fold face: reads the wired faces and folds into
    /// the pure solver (the host owns the sim, this face only reads;
    /// null faces fold to the honest zeros face).</summary>
    public static ShareCard BuildSpec(HospitalRank rank, HospitalAlbum album, HospitalDoctors doctors)
    {
        var rates = new int[3];
        if (album != null)
            for (int i = 0; i < 3; i++) rates[i] = album.RatePctOf(HospitalAlbum.VolumeIds()[i]);
        return SolveSpec(
            rank != null ? rank.Rank : 1,
            rank != null ? rank.ReputationOf() : 0,
            rank != null ? rank.CureRatePctOf() : 0,
            rank != null ? rank.DeptsOf() : 0,
            doctors != null ? doctors.CollecteCount : 0,
            rates);
    }
}

```

```

    /// <summary>Shares the certificate: builds the spec, composes
    /// the PNG through the shared RT face and books evt_album_share
    /// only on a delivered card (Ok=false composes book nothing).
    /// Returns the compose verdict for the caller to surface.</summary>
    public ComposeResult Share(HospitalRank rank, HospitalAlbum album, HospitalDoctors doctors,
        int maxPngBytes = ShareCardService.MaxPngBytes)
    {
        var card = BuildSpec(rank, album, doctors);
        var res = new ShareCardService().Compose(card, maxPngBytes);
        if (res != null && res.Ok)
            _funnel.Add(EventId + "@r" + (rank != null ? rank.Rank : 1) + ":" + res.PngBytes);
        return res;
    }
    /// <summary>Canonical fingerprint: the zero-RNG parity
    /// instrument (same script = same bytes).</summary>
    public string Fingerprint()
    {
        var sb = new System.Text.StringBuilder();
        sb.Append("fun=").Append(_funnel.Count);
        for (int i = 0; i < _funnel.Count; i++) sb.Append('|').Append(_funnel[i]);
        return sb.ToString();
    }
}

// HospitalStage.cs
// HospitalStage.cs - the opening beat driver (G09-T005).
// Drives HospitalIntake through the frozen red-line-11 opening script and
// stamps every promise on the sim clock: the facade hook owns t0, the first
// department decision lands inside the 3-8s window, the queue head registers
// half a second later and the first coin fires visual+sfx+number on one
// tick, then the first fictional patient portrait with the absurd diagnosis
// line (the first laugh) and the first review preview card land inside the
// 10s window. Pure C# zero-RNG: the same script always fingerprints
// byte-identical (Redline11Check consumes this face).
using System.Collections.Generic;
namespace Sy.CrazyHospital
{
    public enum HospitalPhase
    {
        Hook,           // the facade + the street queue caption
        ClinicOpen,     // the first department decision made, doors open
        FirstCoin,       // the first registration coin landed
        FirstLaughter,  // the absurd diagnosis portrait fired (the first laugh)
    }
    public sealed class HospitalStage
    {
        // the frozen opening script (red-line-11 beat anchors)
        public const double DecisionWindowStart = 3.0; // T+3s: the doors decision face goes live
        public const double FirstLaughAt = 8.5;       // T+8.5s: patient portrait + absurd diagnosis
        public const double ReviewPreviewAt = 9.5;    // T+9.5s: the first review preview card
    }
}

```

```

readonly HospitalIntake _intake;
readonly HospitalEvents _events;
readonly HospitalRank _rank; // T008: the rank-review gate rides the opening clock
readonly List<(string id, double t)> _stamps = new List<(string, double)>();
double _t;
bool _deptOpen;
double _deptOpenedAt = -1;
bool _firstCoin;
bool _admitBlocked; // the registration attempt refused this call: retry on the next Advance (per-call law)
bool _laughDone;
bool _previewDone;
string _firstPatientDiag = "";
int _firstCoinGoldBefore;
int _firstCoinJumpHc;
HospitalStage(HospitalIntake intake)
{
    _intake = intake;
    _events = new HospitalEvents(intake, 1); // T007: the event engine rides the opening clock (own timeline -
    _rank = new HospitalRank(intake, _events, 1); // T008: the rank gate rides the same clock (identity: no wr
}
/// &summary&The fresh dean: an unlicensed clinic, one bed, three
/// patients already queuing at the street corner.&/summary&
public static HospitalStage NewDean()
    => new HospitalStage(new HospitalIntake(1));
// ---- live faces ----
public double SimTime => _t;
public HospitalPhase Phase
{
    get
    {
        if (_laughDone) return HospitalPhase.FirstLaughter;
        if (_firstCoin) return HospitalPhase.FirstCoin;
        if (_deptOpen) return HospitalPhase.ClinicOpen;
        return HospitalPhase.Hook;
    }
}
public HospitalIntake Intake => _intake;
public HospitalEvents Events => _events;
public HospitalRank Rank => _rank;
public bool ModalActive => false; // zero login wall, zero blocking modal (red line 11)
public bool CanDecide => _deptOpen == false; // the doors decision is live right after the hook
public IReadOnlyList<(string id, double t)> Stamps => _stamps;
/// &summary&First stamp time of an event id (-1 when never fired).&/summary&
public double TimeOf(string id)
{
    for (int i = 0; i < _stamps.Count; i++)
        if (_stamps[i].id == id) return _stamps[i].t;
    return -1;
}
void Stamp(string id)

```

```

{
    _stamps.Add((id, _t));
}

/// <summary>Advances the opening clock: drives the intake economy
/// and fires the scheduled beats (registration admit, first laugh,
/// review preview) on the frozen script. The dt is sliced at the
/// next scheduled beat so every promise lands on its scripted sim
/// time no matter how coarse the caller drives the clock (a whole
/// 13s hop still books the +0.5s admit on time - the bed turn then
/// completes before a review at t+13, the ladder walk rides this).</summary>
public void Advance(double dt)
{
    if (dt <= 0) return;
    if (TimeOf("evt_hospital_hook") < 0)
        Stamp("evt_hospital_hook"); // the hook owns t0 (facade + street queue + caption)
    _admitBlocked = false; // the registration retry law is per-call (a refused admit waits for the next Advan
    double remaining = dt;
    while (remaining > 0)
    {
        double seg = remaining;
        double next = NextBeatTime();
        if (next - _t < seg) seg = next - _t; // slice at the beat, never past it
        if (seg > 0)
        {
            _t += seg;
            _intake.Advance(seg);
            _events.Advance(seg);
            _rank.Advance(seg); // T008: the reputation-band face tracks the live rep (own funnel, never write
            remaining -= seg;
        }
        FireScheduledBeats();
        if (seg <= 0) remaining = 0; // a due beat just resolved: hand control back to the caller, never sp
    }
}

/// <summary>The earliest pending scheduled beat (double.MaxValue
/// when the script has nothing left to fire). The blocked-admit
/// face keeps a refused registration out of the slice plan until
/// the next Advance call retries it (the old per-call law).</summary>
double NextBeatTime()
{
    double next = double.MaxValue;
    if (_deptOpen && !_firstCoin && !_admitBlocked)
    {
        double admit = _deptOpenedAt + HospitalIntake.FirstAdmitDelaySeconds;
        if (admit < next) next = admit;
    }
    if (_deptOpen && !_laughDone && FirstLaughAt < next) next = FirstLaughAt;
    if (_deptOpen && !_previewDone && ReviewPreviewAt < next) next = ReviewPreviewAt;
    return next;
}

```

```
/// &summary> Fires every scheduled beat whose scripted time the
/// clock has reached.&lt;/summary>
void FireScheduledBeats()
{
    // the queue head steps in half a second after the doors open
    if (_deptOpen && !_firstCoin && !_admitBlocked
        && _t >= _deptOpenedAt + HospitalIntake.FirstAdmitDelaySeconds)
    {
        _firstCoinGoldBefore = _intake.GoldHc;
        if (_intake.TryRegisterQueueHead())
        {
            _firstCoinJumpHc = _intake.GoldHc - _firstCoinGoldBefore; // snapshot at the stamp tick
            Stamp("evt_first_registration");
            Stamp("coin_visual"); // the coin popup face
            Stamp("coin_sfx"); // the signature coin chime trigger
            Stamp("coin_number"); // the gold counter jump
            Stamp("evt_first_coin");
            _firstCoin = true;
            _firstPatientDiag = HospitalTexts.EnText(HospitalTexts.KeyDiagHiccupBug);
        }
        else _admitBlocked = true; // queue or bed refused: retry on the next Advance call
    }
    // the first laugh: the first fictional patient portrait + the absurd diagnosis line
    if (_deptOpen && !_laughDone && _t >= FirstLaughAt)
    {
        Stamp("evt_first_laugh");
        Stamp("evt_first_patient_portrait");
        Stamp("evt_first_diag_line");
        _laughDone = true;
    }
    // the first review preview card (the level-1 review teaser)
    if (_deptOpen && !_previewDone && _t >= ReviewPreviewAt)
    {
        Stamp("evt_review_preview");
        _previewDone = true;
    }
}
/// &summary> The doors decision: opens the first preset department
/// slot (pick WHAT opens). Returns false while the wallet or the
/// rank chain refuses.&lt;/summary>
public bool OpenFirstDepartment(int choice)
{
    if (_deptOpen) return false;
    if (TimeOf("evt_hospital_hook") &lt; 0) return false; // the hook owns to
    if (!_intake.OpenNextSlot(choice)) return false;
    _deptOpen = true;
    _deptOpenedAt = _t;
    Stamp("evt_dept_opened");
    Stamp("evt_first_decision");
    return true;
}
```

```

    }
    public int FirstCoinJumpHc => _firstCoin ? _firstCoinJumpHc : 0;
    public string FirstPatientDiag => _firstPatientDiag;
    /// <summary>Requests the rank review (T008 host face): the engine
    /// owns the verdict and both host folds (the economy row swap and
    /// the stage pool widen land inside the settle). The opening script
    /// never calls this - the first review is a player action, so the
    /// signature law holds.</summary>
    public ReviewVerdict RequestReview() => _rank.RequestReview();
    /// <summary>Opening fingerprint: every stamp, the wallet and the
    /// intake state - byte-identical across identical scripts.</summary>
    public string OpeningSig()
    {
        var sb = new System.Text.StringBuilder();
        foreach (var s in _stamps) sb.Append(s.id).Append('@').Append(s.t.ToString("0.###")).Append(';');
        sb.Append(_intake.Fingerprint());
        return sb.ToString();
    }
}
}
// HospitalTexts.cs
// HospitalTexts.cs - the hospital copy law for Crazy Hospital (G09-T004).
// English source strings live here as compiled code (the family source law:
// code strings stay English); the zh-CN localization rides a separate data
// file (Assets/_Game/Configs/Texts_zhCN.json - data, not source). Lookup
// follows the shared LocaleService contract: a missing key falls back to the
// key itself (machine-readable, never a crash), and a zh entry missing from
// the data file falls back to the English source. Zero scattered copy: every
// UI-facing sentence routes through these keys - DataSchemaCheck scans the
// runtime tree for stray CJK and proves the coverage law. The medical
// disclaimer is a permanent resident key (compliance face, always on).
using System;
using System.Collections.Generic;
using Sy.Core;
namespace Sy.CrazyHospital
{
    public static class HospitalTexts
    {
        // ---- key constants (the compiled copy surface) ----
        public const string KeyHookCaption = "hook_caption";
        public const string KeyMedicalDisclaimer = "medical_disclaimer"; // the permanent compliance face
        public const string KeyRank1 = "rank_1";
        public const string KeyRank2 = "rank_2";
        public const string KeyRank3 = "rank_3";
        public const string KeyRank4 = "rank_4";
        public const string KeyRank5 = "rank_5";
        public const string KeyStatPatients = "stat_patients";
        public const string KeyStatReputation = "stat_reputation";
        public const string KeyStatCureRate = "stat_cure_rate";
        public const string KeyStatGold = "stat_gold";
    }
}

```

```
public const string KeyStatDepts = "stat_depts";
public const string KeyDeptOpen = "dept_open";
public const string KeyDeptSlots = "dept_slots";
public const string KeyRegFee = "reg_fee";
public const string KeyBedFee = "bed_fee";
public const string KeyRepBad = "rep_bad";
public const string KeyRepPlain = "rep_plain";
public const string KeyRepGood = "rep_good";
public const string KeyRepBanner = "rep_banner";
public const string KeyReviewLevel1 = "review_level1";
public const string KeyReviewLevel2 = "review_level2";
public const string KeyReviewLevel3 = "review_level3";
public const string KeyReviewTop = "review_top";
public const string KeyGatePromote = "gate_promote";
public const string KeyGateRectify = "gate_rectify";
public const string KeyResearchTitle = "research_title";
public const string KeyDrugHiccupCola = "drug_hiccup_cola";
public const string KeyDrugSleepwalkPatch = "drug_sleepwalk_patch";
public const string KeyDrugShockSugar = "drug_shock_sugar";
public const string KeyDisHiccupBug = "disease_hiccup_bug";
public const string KeyDisGummyBone = "disease_gummy_bone";
public const string KeyDisMonday = "disease_monday";
public const string KeyDisHeadGrass = "disease_head_grass";
public const string KeyDisCactusAccident = "disease_cactus_accident";
public const string KeyDiagHiccupBug = "diag_hiccup_bug";
public const string KeyDiagGummyBone = "diag_gummy_bone";
public const string KeyDiagMonday = "diag_monday";
public const string KeyDiagHeadGrass = "diag_head_grass";
public const string KeyDiagCactusAccident = "diag_cactus_accident";
public const string KeyAlbumTitle = "album_title";
public const string KeyOfflineSummary = "offline_summary";
public const string KeyOfflineDouble = "offline_double";
public const string KeyNightTide = "night_tide";
public const string KeyCarryLabel = "carry_label";
public const string KeyNewJourney = "new_journey";
public const string KeyNewJourneyConfirm = "new_journey_confirm";
public const string KeyShareChallenge = "share_challenge";
public const string KeyDailyReport = "daily_report";
// ---- the quack-doctor hiring face (T006, SYS03) ----
public const string KeyDoctorDraw = "doctor_draw";
public const string KeyDoctorOnStaff = "doctor_on_staff";
public const string KeyDutyRoster = "duty_roster";
public const string KeyDocThreesec = "doc_threesec";
public const string KeyDocRemedyQuack = "doc_remedy_quack";
public const string KeyDocMoyu = "doc_moyu";
public const string KeyDocWangdali = "doc_wangdali";
public const string KeyDocNightNurse = "doc_night_nurse";
public const string KeyDocBeautyChief = "doc_beauty_chief";
public const string KeyQuirkThreesec = "quirk_threesec";
public const string KeyQuirkRemedyQuack = "quirk_remedy_quack";
```

```

public const string KeyQuirkMoyu = "quirk_moyu";
public const string KeyQuirkWangdali = "quirk_wangdali";
public const string KeyQuirkNightNurse = "quirk_night_nurse";
public const string KeyQuirkBeautyChief = "quirk_beauty_chief";
// ---- the event-engine face (T007, SYS04): card titles, the
// inspection preview and the two choice labels ----
public const string KeyEventPreview = "event_preview";
public const string KeyEventChoiceGo = "event_choice_go";
public const string KeyEventChoiceSlack = "event_choice_slack";
public const string KeyEventE01 = "event_e01";
public const string KeyEventE02 = "event_e02";
public const string KeyEventE03 = "event_e03";
public const string KeyEventE04 = "event_e04";
public const string KeyEventE05 = "event_e05";
public const string KeyEventE06 = "event_e06";
public const string KeyEventE07 = "event_e07";
public const string KeyEventE08 = "event_e08";
public const string KeyEventE09 = "event_e09";
public const string KeyEventE10 = "event_e10";
// ---- the rank-review face (T008, SYS05/SYS06): the promotion
// ceremony show and the rectification comedy show beats ----
public const string KeyCeremonyShow = "ceremony_show";
public const string KeyRectifyShow = "rectify_show";
/// &summary&English source table (compiled - the source-of-law face).&/summary&
public static readonly Dictionary&string, string& En = new Dictionary&string, string&
{
    { KeyHookCaption, "WELCOME TO CRAZY HOSPITAL - YOUR DEAN LIFE STARTS NOW" },
    { KeyMedicalDisclaimer, "FOR ENTERTAINMENT ONLY - NOT MEDICAL ADVICE" },
    { KeyRank1, "UNLICENSED CLINIC" },
    { KeyRank2, "LEVEL-1 HOSPITAL" },
    { KeyRank3, "LEVEL-2 HOSPITAL" },
    { KeyRank4, "LEVEL-3 HOSPITAL" },
    { KeyRank5, "TOP-GRADE STAMPED" },
    { KeyStatPatients, "PATIENTS" },
    { KeyStatReputation, "REPUTATION" },
    { KeyStatCureRate, "CURE RATE" },
    { KeyStatGold, "GOLD" },
    { KeyStatDepts, "DEPTS" },
    { KeyDeptOpen, "OPEN A DEPARTMENT" },
    { KeyDeptSlots, "SLOTS: {0}/{1}" },
    { KeyRegFee, "REG FEE" },
    { KeyBedFee, "BED FEE" },
    { KeyRepBad, "REVIEW STORM - THE STREETS GO QUIET" },
    { KeyRepPlain, "QUIET STREETS, BUSINESS AS USUAL" },
    { KeyRepGood, "STEADY PATIENTS, STEADY LINES" },
    { KeyRepBanner, "BANNER-WALL QUEUES OUT THE DOOR" },
    { KeyReviewLevel1, "LEVEL-1 REVIEW (THE PAPERWORK LITERATURE PRIZE)" },
    { KeyReviewLevel2, "LEVEL-2 REVIEW (THE MYSTERY PATIENT STRIKES)" },
    { KeyReviewLevel3, "LEVEL-3 REVIEW (THE GREAT HYGIENE INSPECTION)" },
    { KeyReviewTop, "TOP-GRADE REVIEW (THE REHEARSED MIRACLE SHOW)" },

```

```

{ KeyGatePromote, "PROMOTED! {0} UNLOCKED" },
{ KeyGateRectify, "RECTIFY AND RESUBMIT - NEXT REVIEW WINDOW SOON" },
{ KeyResearchTitle, "WONDER-DRUG LAB" },
{ KeyDrugHiccupCola, "HICCUP COLA - THE BURP GETS LOUDER" },
{ KeyDrugSleepwalkPatch, "SLEEPWALK PATCH - WALK MORE, DREAM DEEPER" },
{ KeyDrugShockSugar, "SHOCK SUGAR - FULL-BODY PIXEL JITTER" },
{ KeyDisHiccupBug, "THE HICCUP BUG" },
{ KeyDisGummyBone, "GUMMY-BONE SYNDROME" },
{ KeyDisMonday, "MONDAY SYNDROME" },
{ KeyDisHeadGrass, "GRASS ON THE HEAD (CODER EDITION)" },
{ KeyDisCactusAccident, "CACTUS-DIET ACCIDENT" },
{ KeyDiagHiccupBug, "DIAGNOSIS: PRESCRIBE ONE MORE HICCUP COLA." },
{ KeyDiagGummyBone, "DIAGNOSIS: BONE IS NOW DESSERT. AVOID DESSERT." },
{ KeyDiagMonday, "DIAGNOSIS: SEVERE MONDAY. COME BACK ON FRIDAY." },
{ KeyDiagHeadGrass, "DIAGNOSIS: WATER TWICE A DAY, NO MOWING." },
{ KeyDiagCactusAccident, "DIAGNOSIS: THE CACTUS IS FINE. THE PATIENT IS NOT." },
{ KeyAlbumTitle, "FICTIONAL DISEASE ALBUM" },
{ KeyOfflineSummary, "WHILE YOU WERE GONE..." },
{ KeyOfflineDouble, "DOUBLE IT?" },
{ KeyNightTide, "NIGHT EMERGENCY TIDE!" },
{ KeyCarryLabel, "NEXT-DEAN CARRY: {0}" },
{ KeyNewJourney, "START A NEW DEAN LIFE" },
{ KeyNewJourneyConfirm, "TAP AGAIN - THE CARRY SEEDS YOUR NEXT CLINIC" },
{ KeyShareChallenge, "CAN YOUR HOSPITAL BEAT MINE?" },
{ KeyDailyReport, "DAY {0}: {1}" },
// ---- the quack-doctor hiring face (T006): names, mischief beats
// and the spree lines - pure comedy, zero medical claims ----
{ KeyDoctorDraw, "HIRE 10 DOCTORS (AD-FUNDED)" },
{ KeyDoctorOnStaff, "ALREADY ON PAYROLL - THE JOKE HIRES TWICE" },
{ KeyDutyRoster, "ON DUTY" },
{ KeyDocThreesec, "THREE-SECOND DOCTOR" },
{ KeyDocRemedyQuack, "FOLK-REMEDY QUACK" },
{ KeyDocMoyu, "SLACKING DIRECTOR" },
{ KeyDocWangdali, "WANG-DALI THE BONE BOLTER" },
{ KeyDocNightNurse, "NIGHT-SHIFT NURSE" },
{ KeyDocBeautyChief, "BEAUTY-INSTITUTE CHIEF" },
{ KeyQuirkThreesec, "NEXT! - THE THREE-SECOND CONSULT, QUEUE MOVES ITSELF" },
{ KeyQuirkRemedyQuack, "ANCESTRAL REMEDY, MYSTICAL EFFECTS - NO GUARANTEES, OBVIOUSLY" },
{ KeyQuirkMoyu, "THE VANISHING WARD ROUND - NOBODY SAW HIM ALL DAY" },
{ KeyQuirkWangdali, "BONES BOLTED ON LIKE SCREWS - DO NOT ASK" },
{ KeyQuirkNightNurse, "SLEEPWALKS THE FLAG PATROL AT 3 A.M. - NEVER TIRED, EVER" },
{ KeyQuirkBeautyChief, "WANTED A STAR FACE, GOT A PIXEL FACE" },
// ---- the event-engine face (T007): the comedy lives in the
// titles - five inspection cards + five daily cards ----
{ KeyEventPreview, "THE INSPECTORS ARE COMING - EVERYONE LOOK BUSY!" },
{ KeyEventChoiceGo, "PLAY IT STRAIGHT" },
{ KeyEventChoiceSlack, "WING IT" },
{ KeyEventE01, "SURPRISE HYGIENE INSPECTION (EVERYONE PRETEND TO WORK)" },
{ KeyEventE02, "THE LEADERSHIP VISIT (THE REHEARSED MIRACLE SHOW)" },
{ KeyEventE03, "THE MYSTERY PATIENT (A SERVICE-ATTITUDE EXAM)" },

```

```

    { KeyEventE04, "REVIEW-PAPER LITERATURE PRIZE (NEGATIVE-GROWTH OPTIMIZATION)" },
    { KeyEventE05, "FIRE-DRILL STRETCHER RELAY (WANG-DALI CARRIES THREE)" },
    { KeyEventE06, "THE SLEEPWALKING HERO ROUNDS THE WARD AND HANGS A BANNER" },
    { KeyEventE07, "THE HICCUP KID STARTS A HOSPITAL-WIDE HICCUP TIDE" },
    { KeyEventE08, "THE CACTUS REGULAR (A SPINY ATTRIBUTION COMEDY)" },
    { KeyEventE09, "THE OVERTIMED CODER GROWS GRASS" },
    { KeyEventE10, "THE MIRROR PRINCESS ORDERS A STAR FACE (GETS A PIXEL FACE)" },
    // ---- the rank-review face (T008): the two show beats ----
    { KeyCeremonyShow, "THE STAMP CEREMONY! FIREWORKS OVER THE COURTYARD, A RAIN OF BANNERS" },
    { KeyRectifyShow, "QUACK-REMEDY SIDE EFFECT - THE REVIEW EXPERT CHECKS INTO OUR OWN WARD 3" },
};

/// <summary>English lookup (missing key falls back to the key itself
/// - machine-readable, never a crash).</summary>
public static string EnText(string key)
    => En.TryGetValue(key, out var v) && !string.IsNullOrEmpty(v) ? v : key;
/// <summary>Parses the zh-CN data file payload into a key-to-text
/// map (values must be non-empty strings; anything else drops the
/// entry so the lookup degrades to English, never to garbage).</summary>
public static Dictionary<string, string> ParseZh(string json)
{
    var map = new Dictionary<string, string>();
    if (string.IsNullOrEmpty(json)) return map;
    try
    {
        if (!(MiniJson.Deserialize(json) is Dictionary<string, object> d)) return map;
        foreach (var kv in d)
            if (!string.IsNullOrEmpty(kv.Key) && kv.Value is string s && !string.IsNullOrEmpty(s))
                map[kv.Key] = s;
    }
    catch
    {
        // corrupt data file = empty map = English fallback face
        return new Dictionary<string, string>();
    }
    return map;
}
/// <summary>LocaleService-compatible lookup delegate over a parsed
/// zh map: zh-CN hits read the data file, everything else (and any
/// miss) reads the English source. Wire-ready for the T005 UI face.</summary>
public static Func<string, string, string> LookupWithZh(Dictionary<string, string> zh)
{
    return (lang, key) =>
    {
        if (lang == "zh-CN" && zh != null && zh.TryGetValue(key, out var v) && !string.I
            return v;
        return EnText(key);
    };
}
}
}

```